RNGD

Concept-to-Schematic AI Studio

System walkthrough and results

A full run of every mode, with screenshots and the actual results of that run

Run	2026-10-05 15:55:50 to 2026-10-05 16:01:56
Report built	2026-10-11 00:02
App	`src/app.py` (Streamlit)

Semester proof of concept. Pipeline, IFC-studio zoning and BaaP data are synthetic; the IFC file, IBC rule set, ADA standards and hotel program are RNGD-supplied. Not for construction.

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1 Executive summary

This report documents a complete run of the RNGD AI Studio: every mode was driven through the real app in a browser, the way a user would use it, and each screen was captured. The numbers below come from that run.

Mode	Headline result	Detail
Multi-Agent Pipeline	REVISE (93%)	2 conflicts; 11 agent steps
IFC Design Studio	1/5 scenarios pass sample zoning	passing: Right-sized: 60 units, podium (compliant)
Open IFC File	IFC2X3, 1,537 elements	3 storeys; 4 outliers excluded
Hotel Schematic	117 keys, 6 levels	13/16 checks pass; fails: Stories above grade
Layout optimization	192 rooms placed on the 3 QA sites, 0 without a window	reference GA: 171 rooms, 24 without a window; hotel: 740 layouts in 36 s, first choice Bar 265x58 · center core
Automated tests	51/51 pass	

Key findings

- **Pipeline:** (see the data-source note in the pipeline section) the arbiter returned **REVISE**. The building footprint itself is geometrically compliant (no setback, easement, or overlap violations; lot coverage 46.3%), but the design fails to resolve a critical, high-severity parking conflict identified earlier in the pipeline. The site plan delivers only 15 surface parking stalls against a required 150 (1.25/unit), a shortfall of 135 stalls. The critic loop explicitly acknowledged this cannot be fixed by mino...
- **IFC Design Studio:** the Baseline (120 units) fails 4 sample-zoning check(s): FAR, Lot coverage %, Parking stalls, Building inside setbacks/easement. Scenarios passing all sample checks: Right-sized: 60 units, podium (compliant). Against RNGD's real IBC rules the Right-sized scenario still fails: IBC-R1-027 (Dead-end corridor maximum for sprinklered R-1).
- **Real IFC file:** opened and summarised without errors; far-away outliers are excluded so the building fills the view.
- **Hotel:** 117 of 117 keys placed on 5 guest floors; failing checks: Stories above grade (a Type IA podium or one fewer guest floor resolves the story limit). ADA mobility rooms: 7 provided vs $\geq$ 7 for 117 rooms (5 + 2 with roll-in) (PASS).
- **Layout optimization (Section 9):** on the three QA sites the updated optimizer places 192 rooms with 0 windowless guest rooms in 59 s, against 171 rooms with 24 windowless for the reference GA (same program, same evaluator). In Hotel Schematic it generated and ranked 740 layouts in 36 s; the first-ranked one (Bar 265x58 · center core, 4 guest floors) places 117 of 117 keys. No language model is used in the optimization.
- **Quality:** 0 on-page errors during the browser run; 51 of 51 automated tests passed.

2 How the run was done

1. The app was started with `streamlit run src/app.py` on `localhost:8501`.
2. `src/capture_app.py` drove headless Chrome through every mode: ran the full pipeline, generated the Baseline and Right-sized IFC scenarios, opened the real IFC file, used every tab of the hotel workspace, generated the hotel PDF, and asked the assistant a question. It waited for each step to finish before taking a screenshot.
3. This report reads the results those steps saved (`outputs/project_001/project_state.json`, the hotel report manifest), recomputes the deterministic results (IFC scenarios, hotel schematic) and runs the test suite.

Setting	Value
Run date	2026-10-05 15:55:50
Python / Streamlit	3.11.11 / 1.52.2
OS	Windows 10
LLM provider	OpenRouter (Chinese-developed models are blocked in code)
Light model	openai/gpt-oss-20b
Reasoning model	openai/gpt-oss-120b

Setting	Value
Arbiter model	anthropic/claude-sonnet-5
Report-writer model	anthropic/claude-sonnet-5.5
Embeddings	all-MiniLM-L6-v2
Screenshots	31 at 1680 x 1050 px, headless Chrome
On-page errors during the run	0

Step	Time
startup	2.2 s
pipeline run	104.8 s
ifc baseline	3.8 s
ifc rightsized	5.1 s
ifc open	52.4 s
hotel pdf	24.2 s
assistant reply	36.8 s

3 System architecture

Section 4 explains how each design is generated and which technology is used.

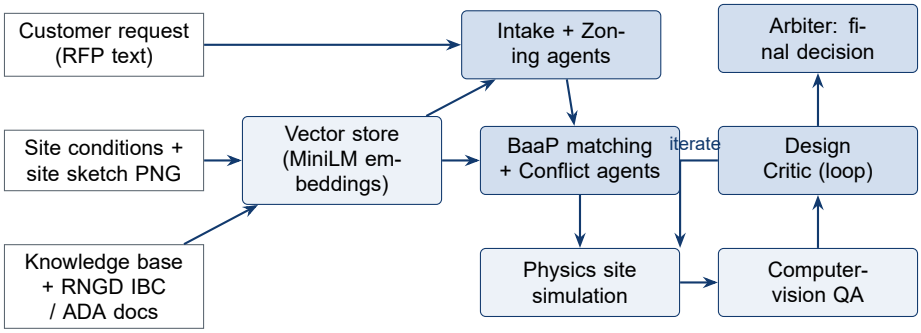


Figure 1: Multi-agent pipeline. Shaded boxes call an LLM through OpenRouter; the others are deterministic Python.

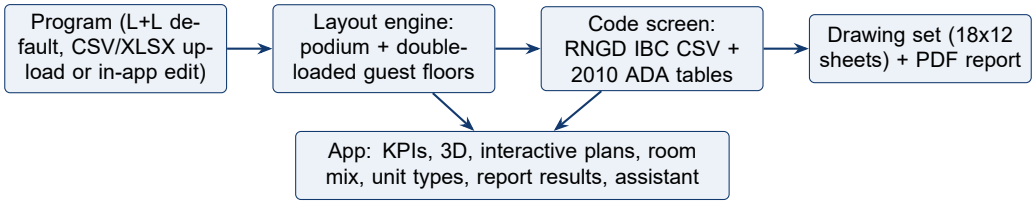


Figure 2: Hotel Schematic mode (RNGD's first use case): deterministic, no LLM needed except the assistant.

4 How the designs were created: methods and technology

Every design in this report is produced by code in this repository. Only the agent steps of the pipeline and the chat assistant use a language model; all geometry (site plans, floor plates, IFC models, hotel plans, drawing sheets and 3D views) is computed deterministically, so the same inputs always give the same design.

4.1 Technology stack

Layer	Technology	Used for
Language	Python 3.11	All engines, agents, drawings and report builders.
User interface	Streamlit 1.52	The five-mode web app, chat panel, program editor and report downloads.
Language models	OpenRouter (OpenAI-compatible SDK)	Agent steps and the assistant. A light model for parsing, a reasoning model for analysis, a Claude model for the single arbiter call. Chinese-developed models are blocked in code; requests time out after 90 s.
Retrieval (RAG)	sentence-transformers all-MiniLM-L6-v2 (TF-IDF fallback)	Vector store over the sample knowledge base plus RNGD's real IBC rule CSV / workbook and the ADA 2010 PDF, tagged synthetic vs real.
Shared state	pydantic models (`ProjectState`)	A blackboard every pipeline agent reads and writes, with a full trace of each step.
Layout optimization	NumPy, SciPy ndimage, scikit-image (least-cost paths), process pool	Architectural decoder, ten-objective fitness, systematic search and memetic GA, plate generator, Pareto + TOPSIS decision (Section 9). No language model in the loop.
Physics simulation	NumPy / SciPy	2D rigid-body relaxation that places the building, driveway and parking inside the buildable envelope.
Computer vision	SciPy ndimage, scikit-image	Reads the site sketch and re-measures the rendered site plan (setbacks, easement, overlaps, parking stalls).
BIM / IFC	IfcOpenShell	Writes IFC4 models (IFC Design Studio) and reads / triangulates real IFC2X3 files (Open IFC File).
Data inputs	pandas, openpyxl, PyMuPDF	Program CSV/XLSX upload, RNGD's IBC rule CSV and workbook, text of the ADA PDF.
Interactive graphics	Plotly	3D models (Mesh3d) and hoverable floor plans in the app.
Drawings	Matplotlib (vector PDF + PNG)	Site plans, floor plates, the 18 x 12 in hotel drawing sheets and the isometric unit / building views.
Documents	LaTeX via Tectonic (XeTeX), PyMuPDF	PDF reports, merged with the drawing sheets.
Verification	pytest, Streamlit AppTest, Playwright + Chrome	Automated tests and the browser run that produced the screenshots in this report.

4.2 Multi-agent pipeline (concept to schematic)

1. **Inputs.** A free-text customer request, `site_conditions.json` and a site sketch PNG with a georeference file.
2. **Vision input (classical CV).** The sketch is thresholded and its lot outline and easement measured in feet using the georeference, then cross-checked against the stated site conditions.
3. **Intake agent (LLM).** Turns the request into a structured program: unit count and mix, floors, amenities, parking preference, budget, hard vs soft requirements. The model must return JSON; malformed JSON is sent back for correction.
4. **Zoning agent (LLM + RAG).** Retrieves the applicable zoning passages and extracts limits (height, stories, FAR, coverage, parking ratio, setbacks). The buildable envelope is then computed in plain Python, not by the model.
5. **BaaP matching agent (LLM + RAG).** Converts the unit mix into exact module counts from the BaaP catalog and estimates gross area and footprint per floor.
6. **Conflict agent (LLM).** Computes feasibility numbers (FAR, coverage, parking demand vs land) and lists conflicts with severities and recommendations.
7. **Physics simulation.** Building, driveway and parking blocks are rigid bodies with containment forces (stay inside the envelope) and separation forces (no overlaps); the system is relaxed until the overlap penalty converges, then rendered as a site plan.
8. **Vision QA.** The rendered plan image is re-measured independently at a known DPI to count stalls and detect setback, easement and overlap violations.

9. **Design critic (LLM) and loop.** Decides whether another simulation pass with adjusted parameters can fix a violation, or whether it is a program / site conflict no geometry change can solve (up to the configured number of iterations).
10. **Floor plate packer (rules).** Lays unit modules along a double-loaded corridor for a ground and a typical floor.
11. **Arbiter (one Claude call).** Reads the whole blackboard and returns APPROVE, APPROVE WITH CONDITIONS, REVISE or INSUFFICIENT INFORMATION with a narrative, unresolved issues and next steps.

4.3 IFC Design Studio (parametric IFC4 model)

1. **Parameters.** Lot, east easement, zoning preset, unit count and mix, floors, wings, parking type (surface or podium), corridor width and floor heights. Five preset scenarios fill these in.
2. **Modules.** The unit mix is turned into module counts from the shared BaaP catalog (`data/baap_catalog.json`), including accessible units, a core, lobby, fitness and BOH/loading.
3. **Layout.** Units are placed on double-loaded corridor wings inside the setbacks and easement. The generator tries one to three wings and keeps the best by score: all units placed, building inside the envelope, a fire lane present, fewest wings.
4. **Parking.** Surface parking is found by scanning the free part of the lot for stall rows; podium parking places stalls under the building as well.
5. **Checks.** Ten sample-zoning checks (units, stories, height, FAR, coverage, parking, setbacks/easement, fire lane, accessible units, exit travel), plus a cross-check of measurable rules from RNGD's real IBC CSV (height, stories, exit travel, dead-end corridor, exits per story).
6. **IFC output.** IfcOpenShell writes a standard IFC4 file with project, site, building and one storey per level, and the unit, core and amenity volumes and slabs on each storey. The file opens in any BIM tool; the app re-reads it for the 3D viewer.

4.4 Open IFC File (real model viewer)

1. IfcOpenShell reads the file's header, project, site, building, storeys and element counts by class.
2. Its geometry iterator triangulates each element (fine curtain-wall framing can be skipped for speed); the meshes are merged per class with NumPy and cached so a second open is fast.
3. The model is recentred on the median of its structural elements; elements more than four building-diagonals (at least 50 m) away are dropped from the view and listed, which is why the real model fills the screen. The original file is never modified.

4.5 Hotel Schematic (RNGD's first use case)

1. **Program.** Default: the Project Renegade Living + Lodging cover-sheet space allocation, row for row (28 rows, 117 keys, 59,350 SF). A CSV/XLSX upload or the in-app editor can replace it; area is the total for each row, and the category (guest, shared, BOH, business, tenant) is read or inferred from the name.
2. **Sorting the program.** Guest rows become keys. Small repeated rooms (Ice, IT, Jan, Supplies) are placed once on every guest floor; small single BOH rooms go into the Level 1 core zone; tenant space becomes a retail block; stairs and corridors are generated, not packed.
3. **Guest floors.** Double-loaded corridor, 6 ft clear; room depth 26 ft and room width = program area / 26 ft. Keys are spread evenly over the floors, largest units first, and alternately to the south and north sides so both ends line up. A 34 ft core at the west end holds Stair 1, two elevators and shafts; Stair 2 is at the east end, so the corridor has an exit at each end. Alternate rooms are mirrored so bathrooms pair up.
4. **Level 1.** The same core stacked below; the lobby first on the street side, other rooms balanced between the two sides; the retail block full depth at the east end, wrapping Stair 2 so stairs stack on every level.
5. **Massing and areas.** Floor-to-floor heights come from the L+L elevations (Level 2 at 16 ft, then 10.75 ft); gross area is drawn length times width per level and is compared with the source sheets and with the meeting rule (33% circulation, 8% wall burden, rounded up to 100 SF).
6. **Code screen.** Rules from RNGD's IBC CSV are evaluated from the geometry: stories, height, area per story, high-rise screen, exits, exit separation (1/3 diagonal), travel distance, corridor width, dead-end corridor and door width, with occupant loads from the CSV's factors. ADA 2010 Tables 224.2, 224.4 and 224.5 set the accessible-room counts. Each failing item gets a suggested fix.

7. **Room detailing.** Each room type is described once in local coordinates (walls with door openings, door swings, bathroom fixtures, beds sized by type, closet, desk, suite partition, spa tub, ADA turning circles) and then transformed into every room on every plan, so plans, enlarged plans and 3D views always agree.
8. **Drawing sheets.** Matplotlib draws vector sheets at true scale in the RNGD L+L format: title block, colour-coded zones, column grid at 30–34 ft bays with bubbles, dimension strings, elevations with an indicative prefab panel breakdown, schedules (gross area, room count, space allocation), enlarged plans and unit 3D views.
9. **3D views.** In the app, Plotly Mesh3d draws every room as a coloured block (hover for details). On the sheets, a small orthographic projector draws boxes with back-face culling and depth sorting (a painter's algorithm), shading faces by orientation; OpenGL was not used.
10. **Report.** LaTeX (Tectonic) builds the front matter and appendices; PyMuPDF merges them with the sheets; a JSON manifest records the results so the app's report tab shows them later.

4.6 Layout optimization (Hotel Schematic "Optimize layouts" and Design Optimizer)

Section 9 gives the architecture, the mathematics and the measured comparison with the reference genetic algorithm. In short: a decoder turns a short genome into a valid plan (straight corridor spines one room depth from the facade, modular bays, rooms assigned to bays), a deterministic fitness scores it, and a systematic search with pattern-search refinement (or a GA) drives the genome. In Hotel Schematic the building shape, the core and stair positions and the floor count are generated as well, and a multi-criteria decision step ranks every distinct layout.

4.7 How this report was made

`src/capture_app.py` drives the running app in headless Chrome with Playwright, waits for each step to finish, takes the screenshots and records timings, on-page errors and the assistant's replies. `src/build_walkthrough_report.py` reads those results, recomputes the deterministic ones, runs or reuses the pytest results and compiles this PDF. `src/build_optimizer_study.py` runs the optimization comparison (reference GA against the updated optimizer, and the hotel layout exploration) and saves its numbers and figures for Section 9.

5 Multi-agent pipeline (project_001, Elm Street Multifamily — sample data)

5.1 What it does

The pipeline turns an unstructured customer request and the site conditions into a Building-as-a-Product (BaaP) schematic: agents parse the program and zoning, match BaaP modules, find conflicts, a physics-style solver places the building and parking, computer vision re-measures the rendered plan, a critic decides whether to iterate, and one arbiter call makes the final decision. The inputs are synthetic sample data.

Customer request (excerpt):

> Subject: Elm Street Multifamily — Program for Concept Design
>
> We'd like RNGD to develop a concept-to-schematic design for our Elm Street site (200 ft of frontage on Elm Street, 300 ft deep, zoned R-4). We are targeting a 4-story, 120-unit market-rate multifamily community.
>
> Unit mix: roughly 20% studios, 45% one-bedrooms, and 35% two-bedrooms. 120 total units is a firm, hard requirement from our lender — please do not go below it, though we could go slightly over if the site allows.
>
> Amenities we need: a resident fitness room, a lobby with an integrated leasing office, a package/mail room, and trash/loading facilities. These are important but their exact size is flexible — treat amenity square footage as a soft preference, not a hard constraint.
>
> Parking: please provide parking per the local code; we'd prefer surface parking to keep costs down if the site allows it, but we're open to you...

5.2 Result

Note. Data source: the tables in this section come from the most recent completed pipeline run on this machine (2026-10-11 00:00, the automated test suite's run through the app). The screenshots come from the browser run that started 2026-10-05 15:55:50. Both runs used the same inputs and code, but LLM outputs differ between runs, so the wording, conflict details and some figures in the screenshots differ from the tables.

Decision	**WARN REVISE** (confidence 93%)
Conflicts found	2 (0 critical, 1 high, 1 medium, 0 low)
Feasible as requested	no
Simulation iterations	1
Agent steps / total time	11 steps, 29.2 s of agent time

Arbiter narrative (LLM output, verbatim). The building footprint itself is geometrically compliant (no setback, easement, or overlap violations; lot coverage 46.3%), but the design fails to resolve a critical, high-severity parking conflict identified earlier in the pipeline. The site plan delivers only 15 surface parking stalls against a required 150 (1.25/unit), a shortfall of 135 stalls. The critic loop explicitly acknowledged this cannot be fixed by minor geometry adjustments and halted iteration without a resolution. This is a program/site capacity conflict that must be resolved before this schematic can be considered a viable concept: either the parking approach must change (structured garage, reduced ratio, off-site/shared parking) or the program/unit count must be adjusted. Approving this as-is would leave a known hard conflict unaddressed. Additionally, the stated local jurisdiction parking ratio has not been confirmed (medium-severity conflict C002), and all outputs are prototype-stage and require licensed professional (architect/civil engineer) review before any entitlement or construction use.

Unresolved issues

- Parking shortfall of 135 stalls (15 detected vs 150 required) — surface parking as requested does not fit on site
- High-severity conflict C001 (surface parking infeasibility) not resolved by end of iteration loop
- Medium-severity conflict C002 (uncertain actual jurisdiction parking ratio) not confirmed
- No structured parking, off-site parking, or program-adjustment alternative has been modeled or costed
- Budget and schedule impact of any parking solution (e.g., parking structure) not yet evaluated against $28M target and accelerated schedule

Recommended next steps

- Confirm actual local zoning parking ratio and minimum required stalls for this unit mix/count
- Model a structured parking option (podium or freestanding garage) or reduced-stall/shared-parking alternative and re-run CV/simulation loop
- Re-evaluate budget impact of structured parking against the $28M construction target
- If stall count cannot be reduced via code or shared parking, revisit unit count/mix to lower parking demand and re-run BaaP module matching
- Route revised schematic through licensed architect/civil engineer and local jurisdiction review prior to any further commitment
- Re-run full pipeline (BaaP matching, conflict detection, simulation/CV QA) once a feasible parking strategy is selected

5.3 Program and site

Item	Value
Units	120
Unit mix	studio 20%, 1br 45%, 2br 35%, 3br 0%
Floors	4
Parking preference	surface parking
Budget	$28,000,000
Hard requirements	120 total units
Lot	200 x 300 ft (60,000 SF)

Item	Value
Setbacks	front 15 ft, rear 20 ft, side_east 10 ft, side_west 10 ft
Buildable envelope	170 x 265 ft (45,050 SF)
Zoning limits	height 65.0 ft, 5 stories, FAR 2.5, coverage 60.0%, parking 1.25/unit
Site sketch cross-check (CV)	consistent

5.4 BaaP plan and conflicts

Modules: STUDIO_A x24, ONEBR_A x54, TWOBR_A x42; unit area 95,190 SF; estimated building 111,988 GSF; footprint 27,997 SF per floor.

ID	Severity	Category	Description	Recommendation
C001	HIGH	parking	Customer prefers surface parking for 150 stalls, but after allocating the building footprint there is only 17,053 sf of land remaining, far short of the 49,500 sf required for surface parking. The site cannot accommodate the requested surface parking configuration.	Shift to a structured (multi-story) parking garage or reduce the number of surface stalls. Evaluate budget impact and schedule implications of adding a parking structure, and verify compliance with ADA accessible-space requirements for the new parking configuration.
C002	medium	zoning/parking ratio	The program calls for 150 parking stalls for 120 residential units (1.25 stalls per unit). If local zoning mandates a lower parking-to-unit ratio, the required stall count may exceed what is permissible.	Confirm the jurisdiction's residential parking ratio. If the ratio is lower, consider reducing total stalls, implementing shared/off-site parking, or providing a reduced-size garage that meets the code-minimum.

5.5 Simulation, vision QA and critic

Iteration	Steps	Max overlap penalty	Converged
1	6	0.438	yes

Pass	Footprint SF	Coverage %	Setback viol.	Easement viol.	Overlaps	Parking	Geometry
1	27,761	46.3	0	0	0	15/150	PASS

- Iteration 1: Parking shortfall of 135 stalls cannot be remedied by minor geometry tweaks; indicates a program/site capacity conflict. (continue: no)

5.6 Agent trace

#	Agent	Action	Model	Status	s	Summary
1	vision_input	parse_site_sketch	classical-cv	ok	0.1	consistent
2	intake	parse_customer_program	openai/gpt-oss-20b	ok	3.7	120 units, 4 floors
3	zoning	analyze_site_and_code	openai/gpt-oss-20b	ok	4.3	buildable envelope 170 x 265 ft (45050 sf)

#	Agent	Action	Model	Status	s	Summary
4	baap_matching	match_program_to_baap	openai/gpt-oss-120b	ok	4.6	120 unit modules, 111988 GSF
5	conflict	check_program_site_baap_conflicts	openai/gpt-oss-120b	ok	4.0	2 conflict(s), feasible_as_requested=False
6	simulation	physics_relax_iteration_1	physics-engine	ok	0.0	6 steps, penalty=0.438, converged=True
7	vision_qa	cv_measure_iteration_1	classical-cv	ok	0.1	geometry_passed=True, stalls 15/150 (shortfall 135), coverage 46.3%
8	design_critic	propose_adjustment_iteration_1	openai/gpt-oss-120b	ok	1.8	continue=False: Parking shortfall of 135 stalls cannot be remedied by minor geometry tweaks; indicates a program/site capacity conflict.
9	floor_plate	pack_typical_upper_floor	rule-based-packer	ok	0.1	14 units placed, 19 deferred
10	floor_plate	pack_ground_floor	rule-based-packer	ok	0.1	11 units placed, 18 deferred (amenities + BOH reserved)
11	arbiter	final_decision	anthropic/claude-sonnet-5	ok	10.4	REVISE (confidence 0.93)

5.7 Assistant note after the run

Not captured.

5.8 Screenshots



Figure 3: Multi-Agent Pipeline before a run: the sample customer request and site sketch that feed the agents; the assistant panel on the right.

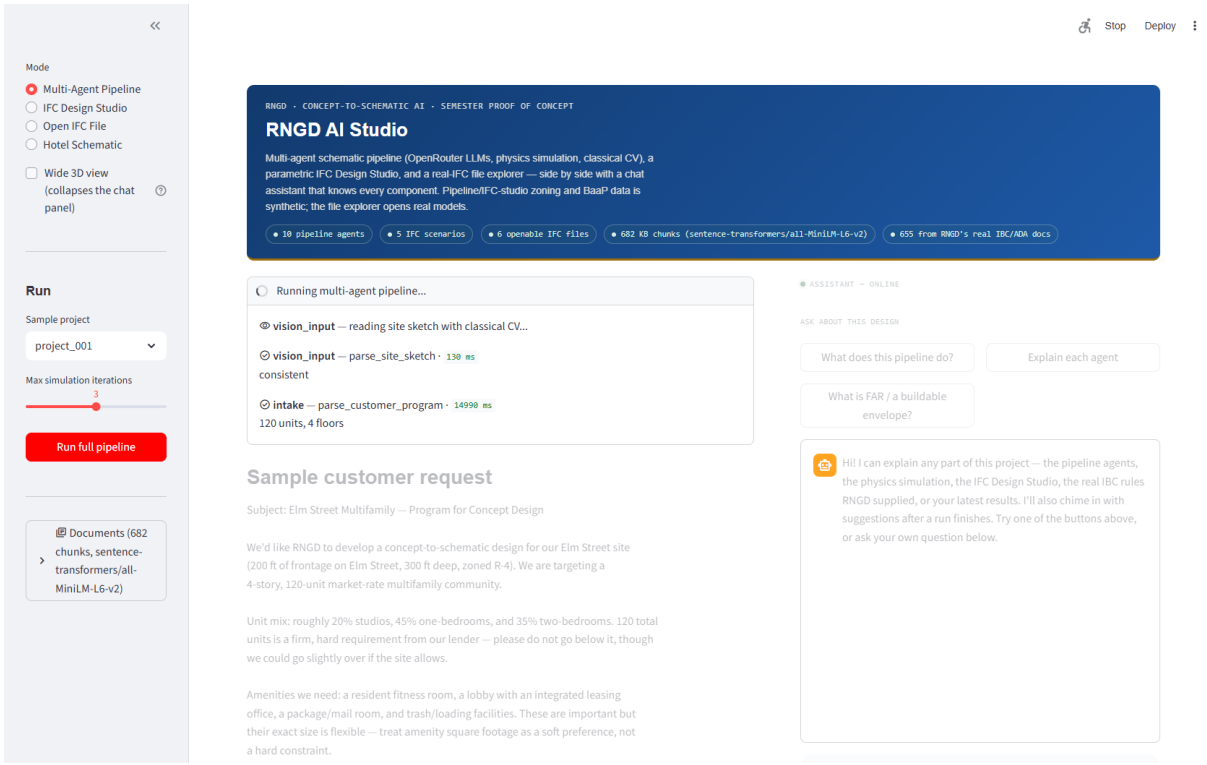


Figure 4: The pipeline running live: each agent reports its step as it finishes.

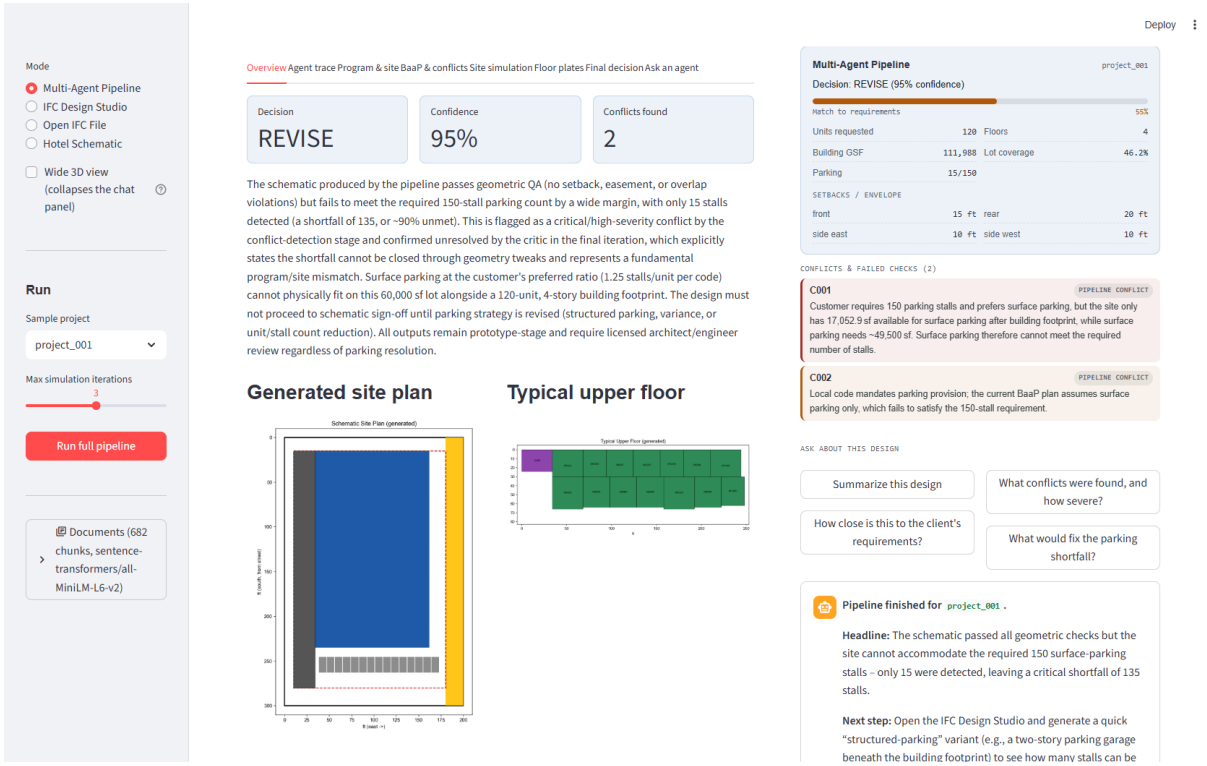


Figure 5: Pipeline result overview: the arbiter's decision, confidence and conflict count.

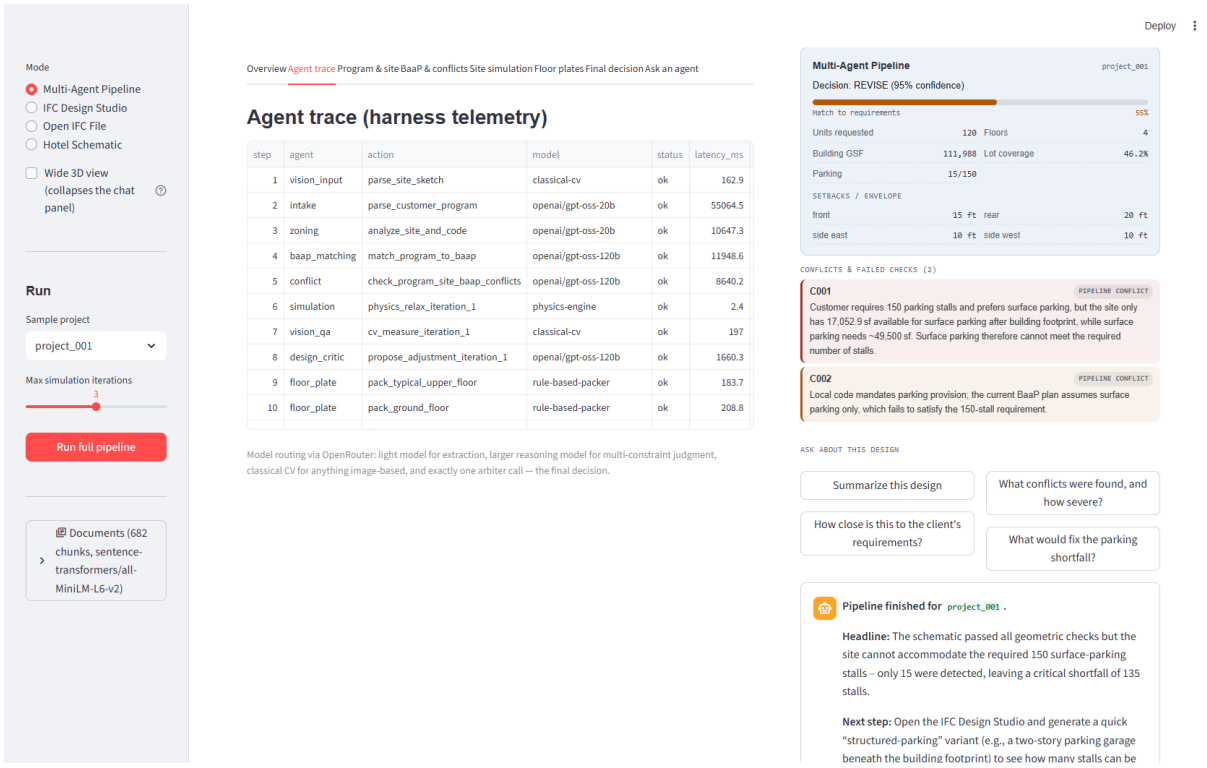


Figure 6: Agent trace: every agent step with model, status and latency.

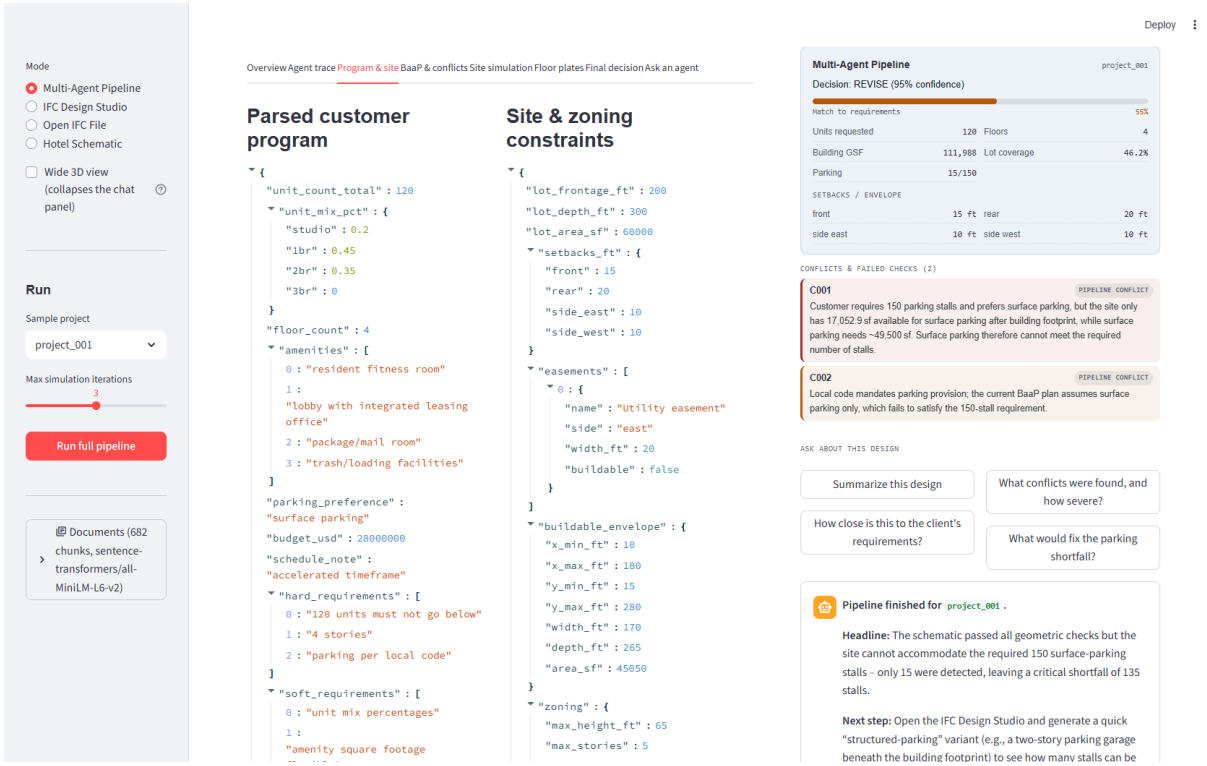


Figure 7: Program parsed from the request and the computed site constraints.

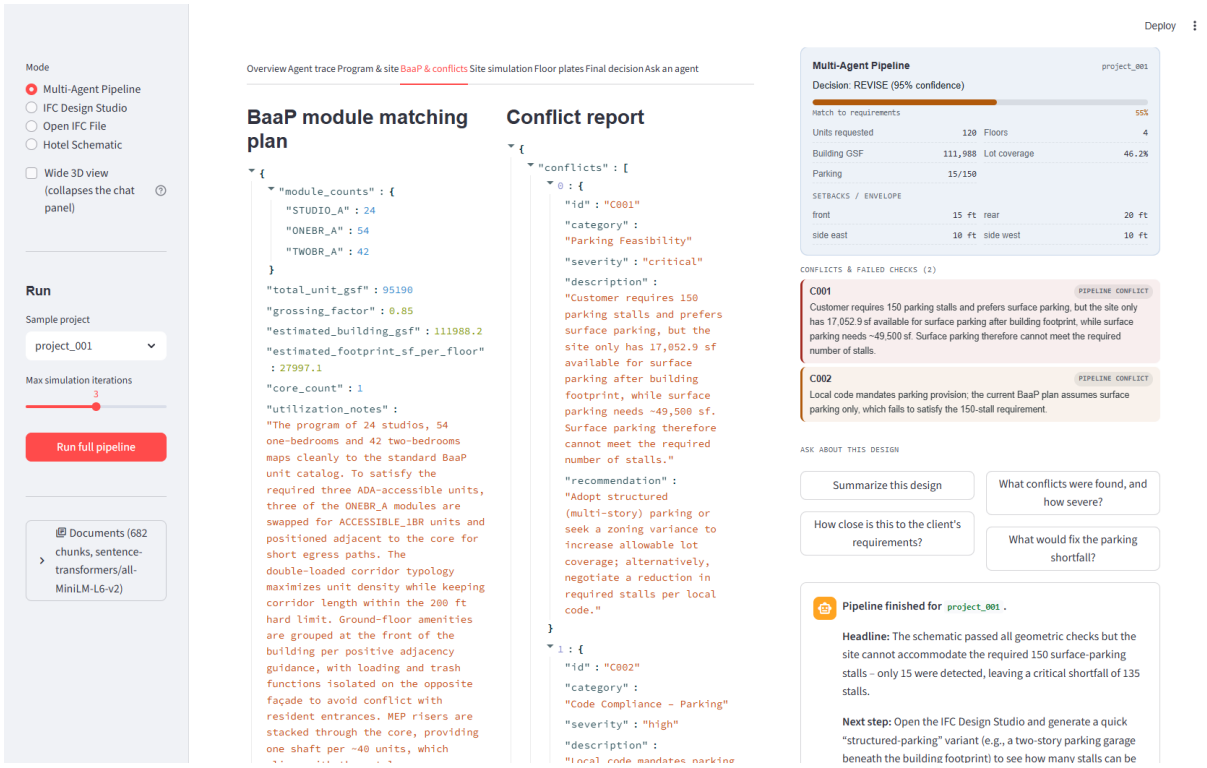


Figure 8: BaaP module counts and the conflicts the Conflict Agent found.

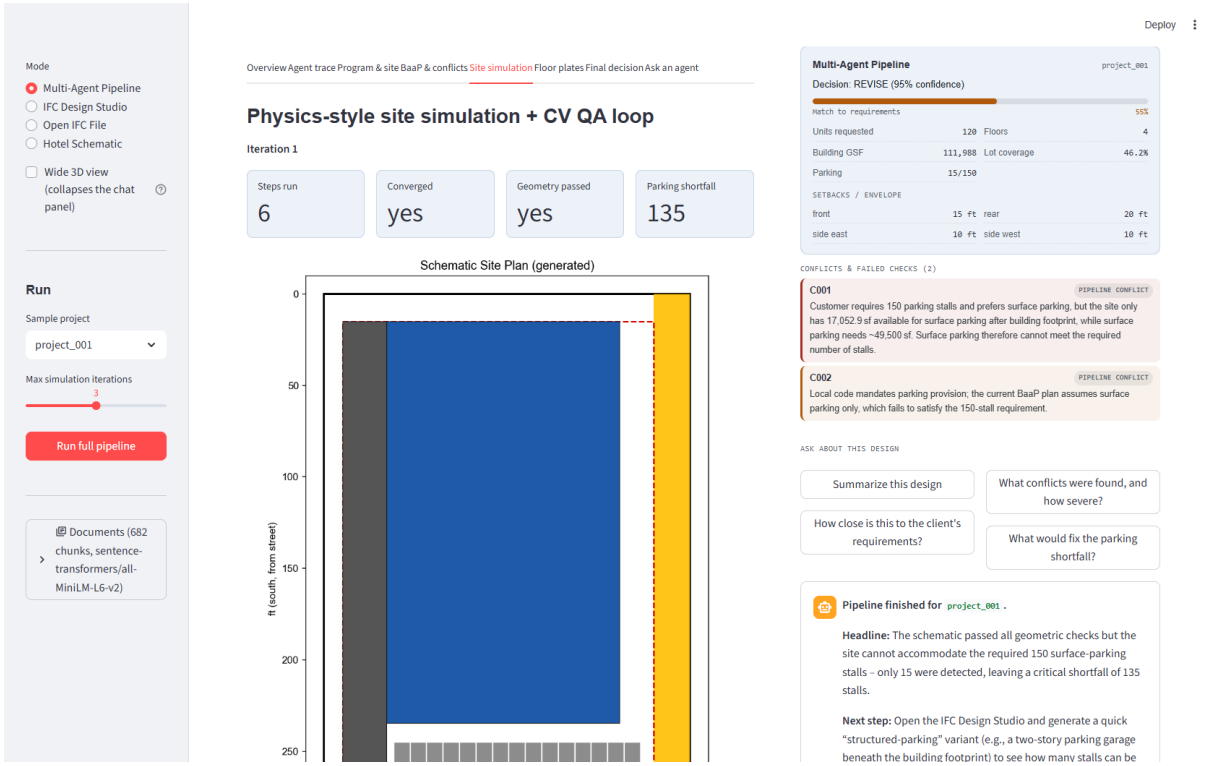


Figure 9: Physics-style site simulation and the computer-vision QA of the rendered plan.

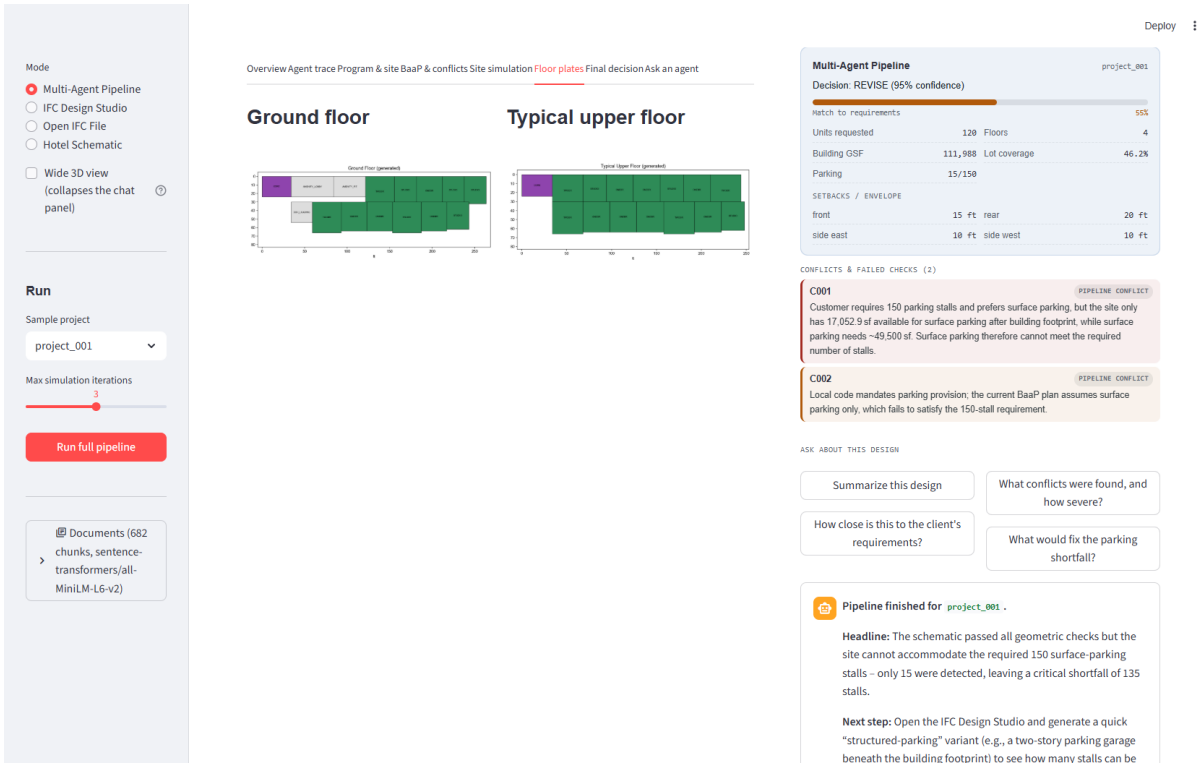


Figure 10: Rule-based floor plates (ground and typical floor).

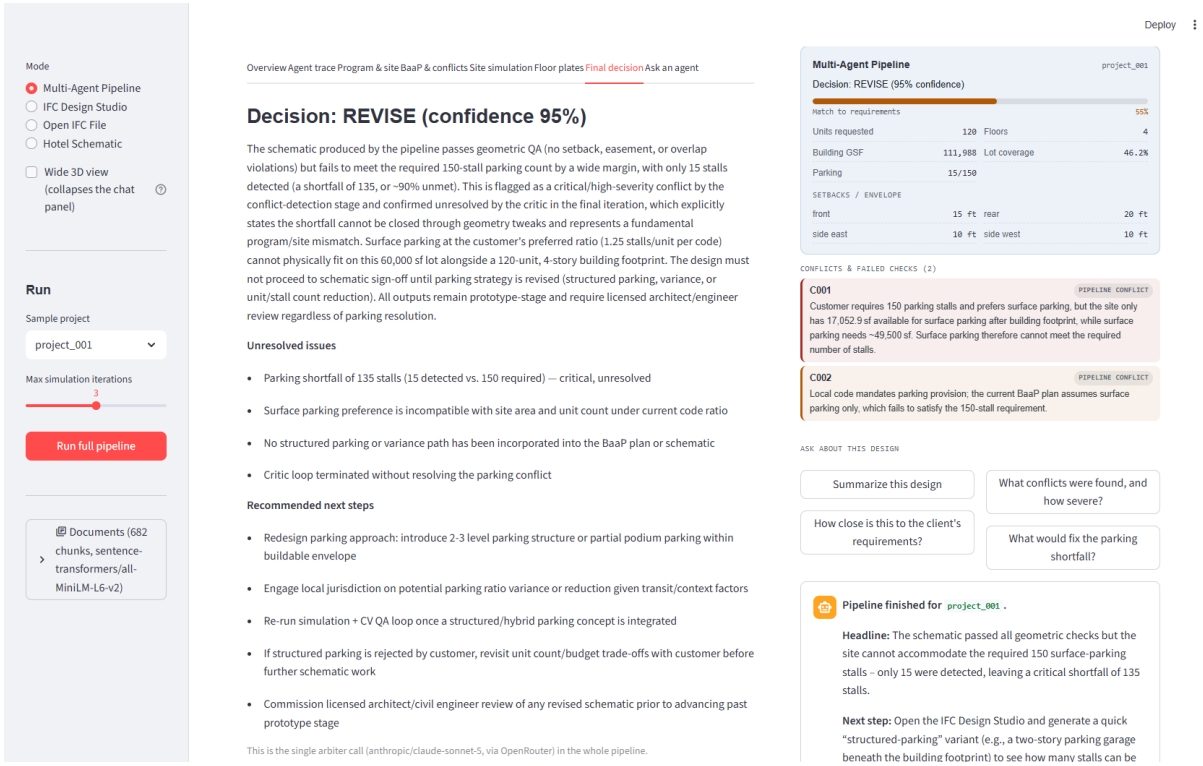


Figure 11: The arbiter's final decision, unresolved issues and next steps.

6 IFC Design Studio (parametric IFC4 generator)

6.1 What it does

A deterministic generator: requirements + a zoning preset + the BaaP catalog produce a building layout, written as a real IFC4 file with IfcOpenShell and shown in an interactive 3D viewer. Every design is scored against ten sample-zoning checks and cross-checked against a subset of RNGD's real 2021 IBC rules. Zoning presets are synthetic.

6.2 All five scenarios

Scenario	Units	Stories	FAR	Coverage	Parking	Sample zoning	Real IBC
Baseline: 4 floors, surface parking	120/120	4	3.16	79.1%	23/150	4 fail	IBC-R1-027
Right-sized: 60 units, podium (compliant)	60/60	5	2.31	46.3%	75/75	**PASS**	IBC-R1-027
Podium parking, 5 floors	120/120	6	3.74	62.4%	120/150	7 fail	IBC-R1-007, IBC-R1-008, IBC-R1-027
Dense R-6, 7 floors + podium	120/120	8	4.01	50.2%	90/90	3 fail	IBC-R1-005, IBC-R1-006, IBC-R1-007, IBC-R1-008, IBC-R1-027
Suburban R-2, 60 units, 3 floors	60/60	3	1.62	54.1%	10/120	3 fail	IBC-R1-027

6.3 Baseline vs right-sized

Side by side: the Baseline (120 units, surface parking) and the Right-sized scenario (60 units, podium parking) that the app offers as the fix.

Check	Required	Baseline		Right-sized	
Units placed	120	120	**PASS**	60	**PASS**
Stories	≤ 5	4	**PASS**	5	**PASS**
Height (ft)	≤ 65	45.5	**PASS**	56.0	**PASS**
FAR	≤ 2.5	3.16	**FAIL**	2.31	**PASS**
Lot coverage %	≤ 60	79.1	**FAIL**	46.3	**PASS**
Parking stalls	150	23	**FAIL**	75	**PASS**
Building inside setbacks/easement	yes	no	**FAIL**	yes	**PASS**
Fire lane / driveway (24 ft)	yes	yes	**PASS**	yes	**PASS**
Accessible units (2%)	3	3	**PASS**	2	**PASS**
Exit travel to core (ft)	≤ 200	180	**PASS**	154	**PASS**

6.4 Baseline against RNGD's real IBC rules

Rule	Rule name	Required	Measured	Status
IBC-R1-005	R occupancy sprinklered Type IIA allowable height	$\leq$ 85 ft	45.5	PASS
IBC-R1-006	R occupancy sprinklered Type IIB allowable height	$\leq$ 75 ft	45.5	PASS
IBC-R1-007	R-1 sprinklered Type IIA allowable stories	$\leq$ 5 stories above grade plane	4	PASS
IBC-R1-008	R-1 sprinklered Type IIB allowable stories	$\leq$ 5 stories above grade plane	4	PASS
IBC-R1-023	R exit access travel distance, sprinklered	$\leq$ 250 ft	180	PASS
IBC-R1-027	Dead-end corridor maximum for sprinklered R-1	$\leq$ 50 ft	180	FAIL
IBC-R1-015	Minimum exits per story: 1-500 occupants	2 exits (occupant load 1-500)	6 exits (3 core(s) x 2 stairs), ~140 occupants/floor	PASS

6.5 Screenshots

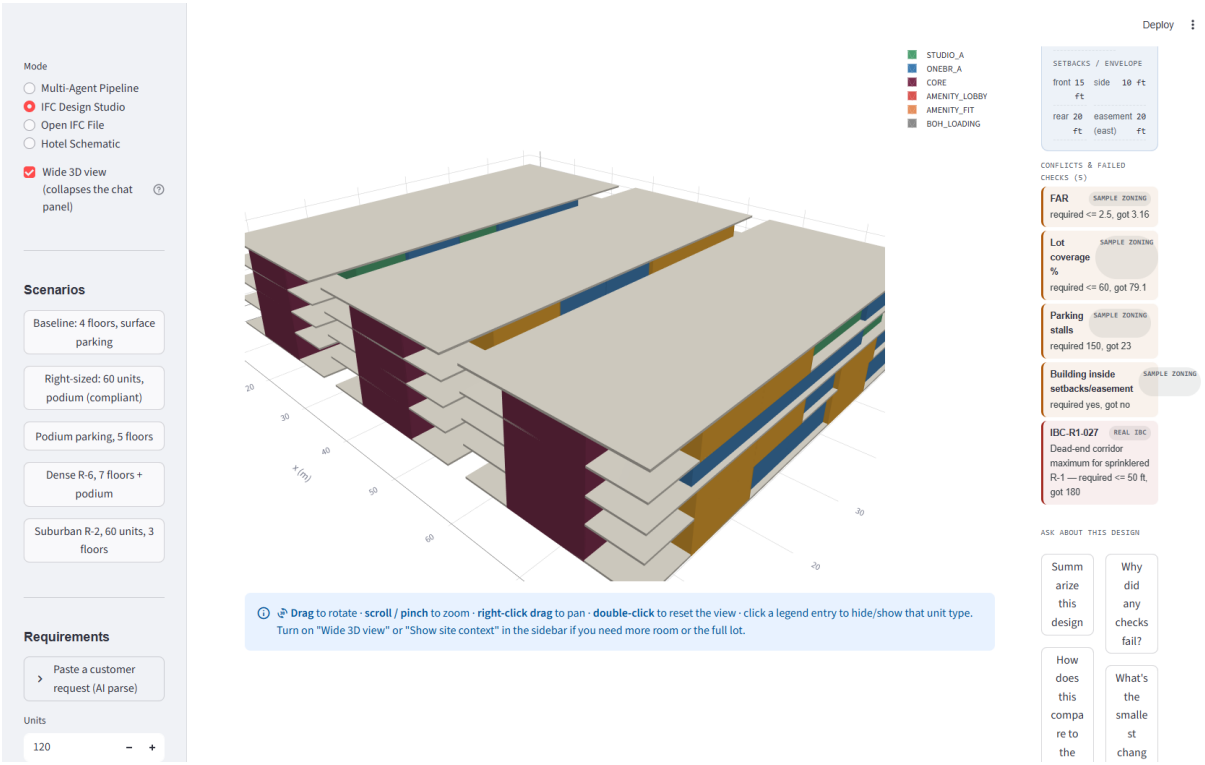


Figure 12: IFC Design Studio, Baseline scenario (120 units, 4 floors, surface parking): the generated IFC4 model in the 3D viewer.

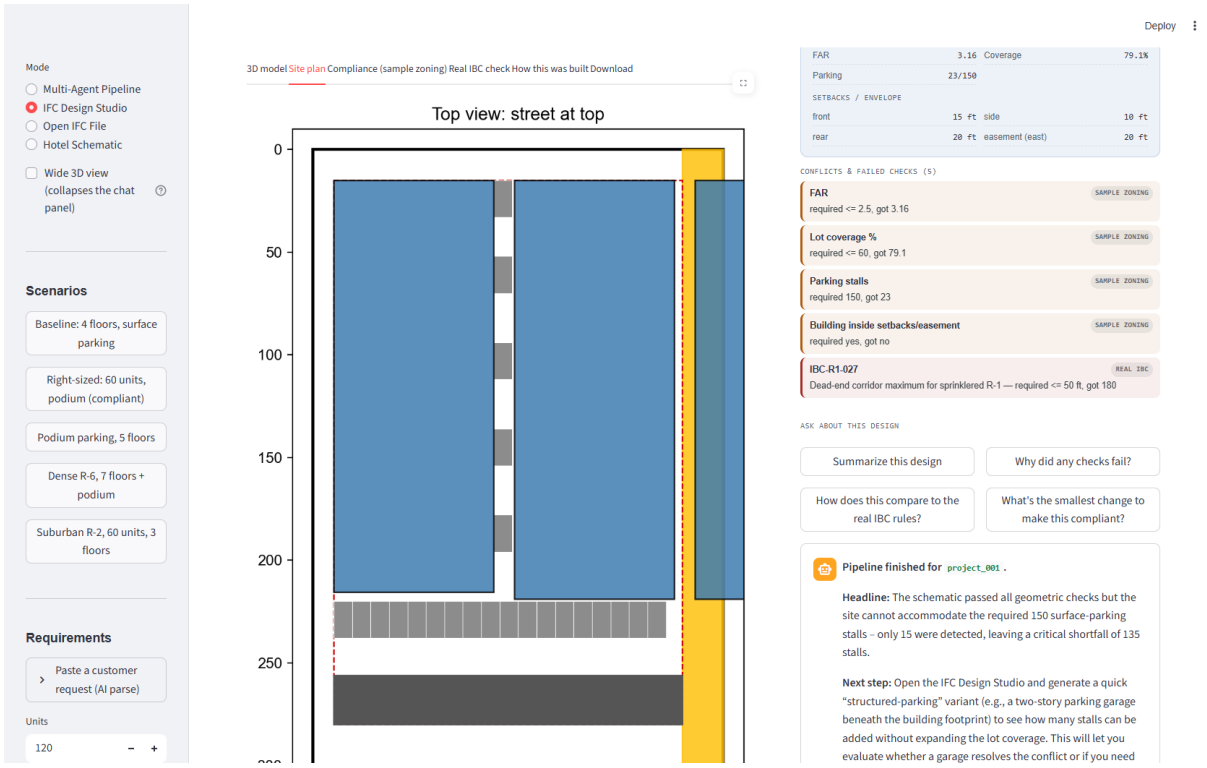


Figure 13: Baseline site plan: building, parking, easement and fire lane on the synthetic lot.

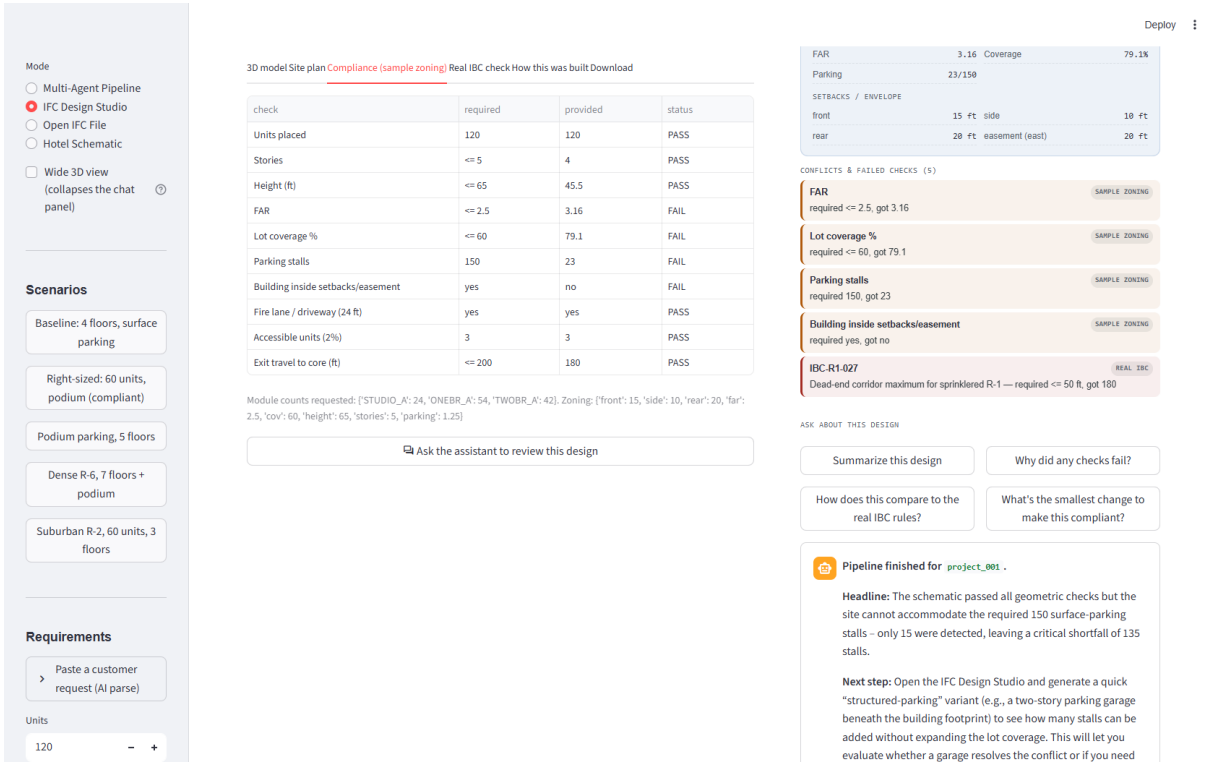


Figure 14: Baseline compliance against the ten sample-zoning checks.

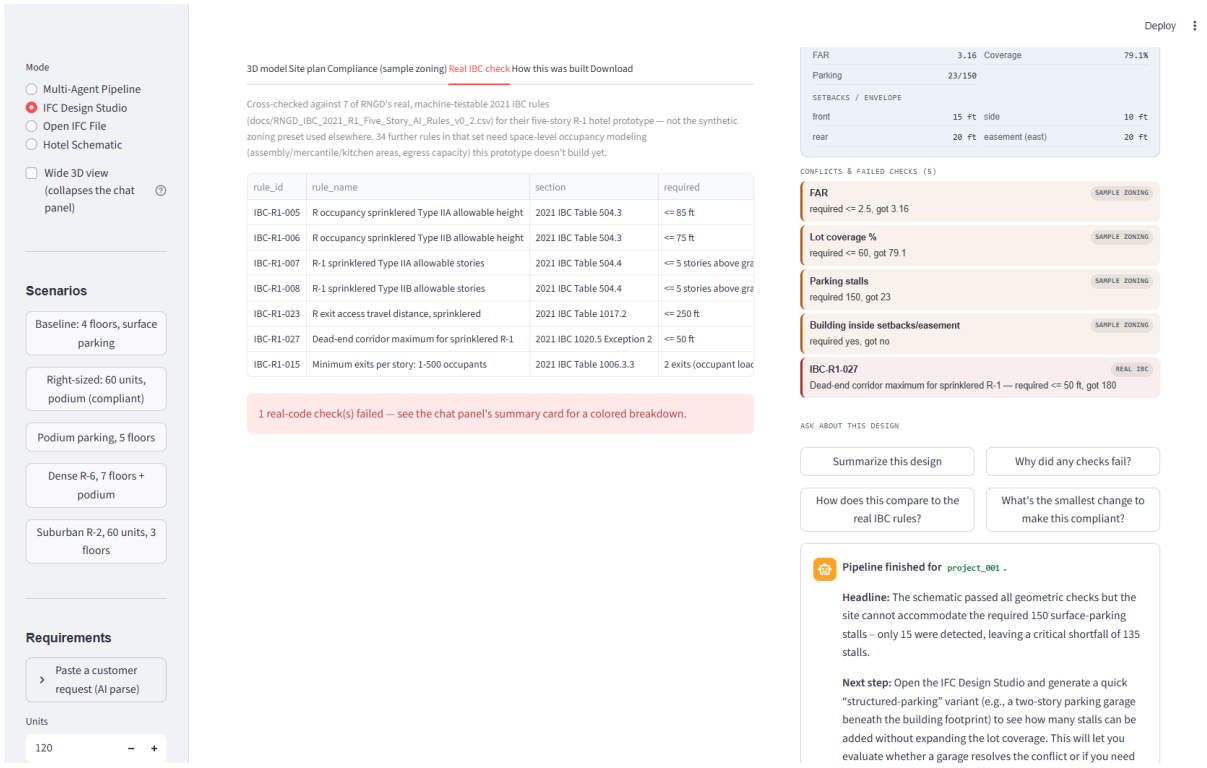


Figure 15: Baseline cross-checked against RNGD's real 2021 IBC rules.

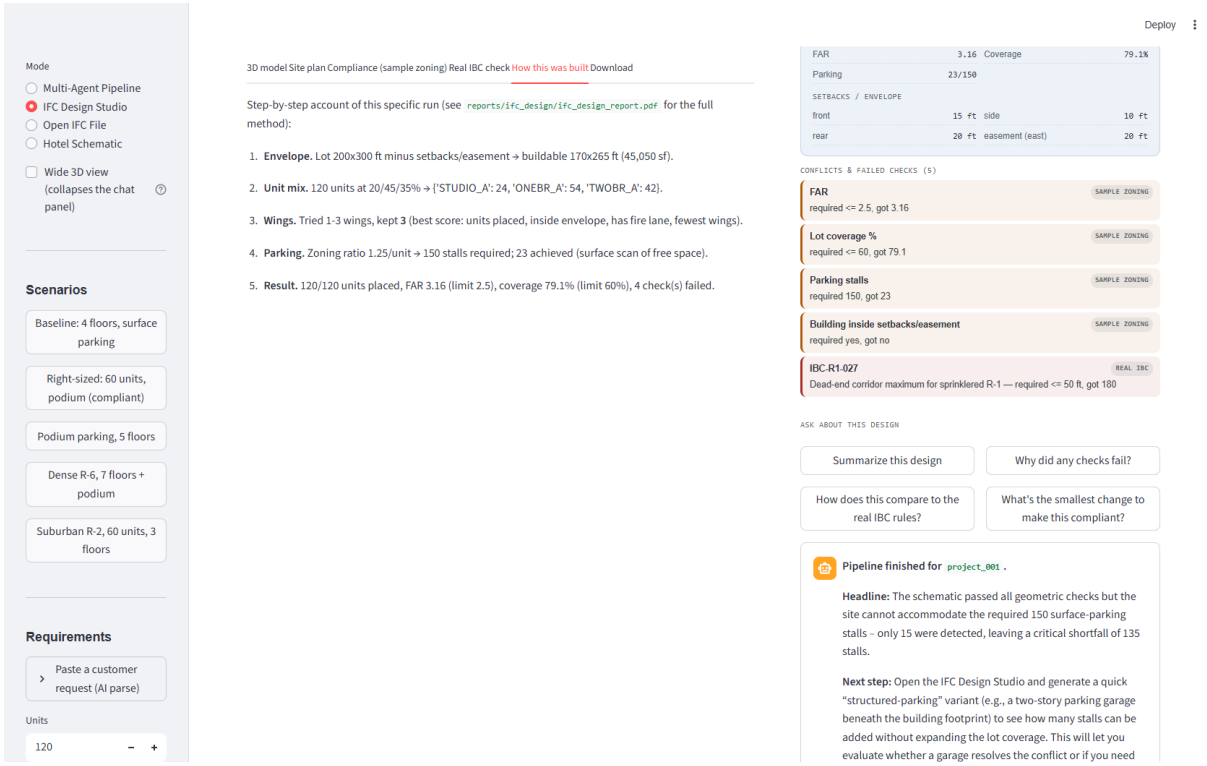


Figure 16: Step-by-step explanation of how the layout was generated.

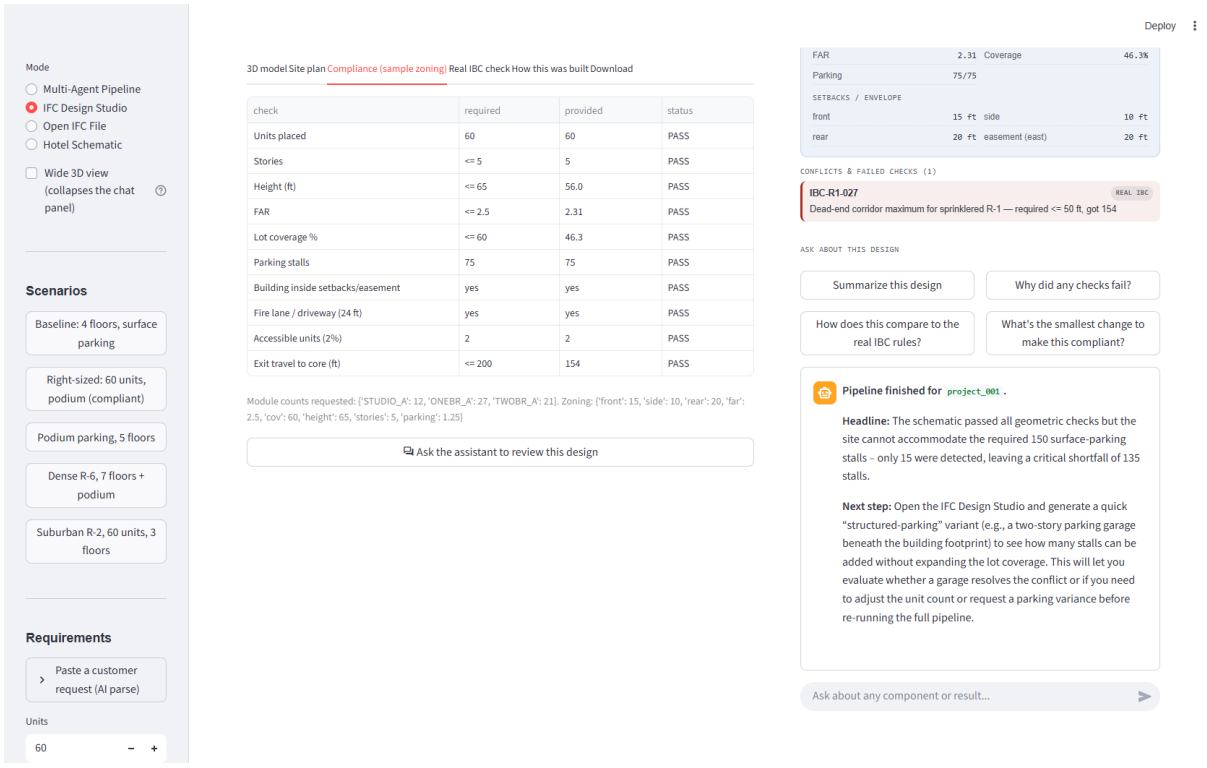


Figure 17: Right-sized scenario (60 units, podium parking): the compliance checks after the fix.

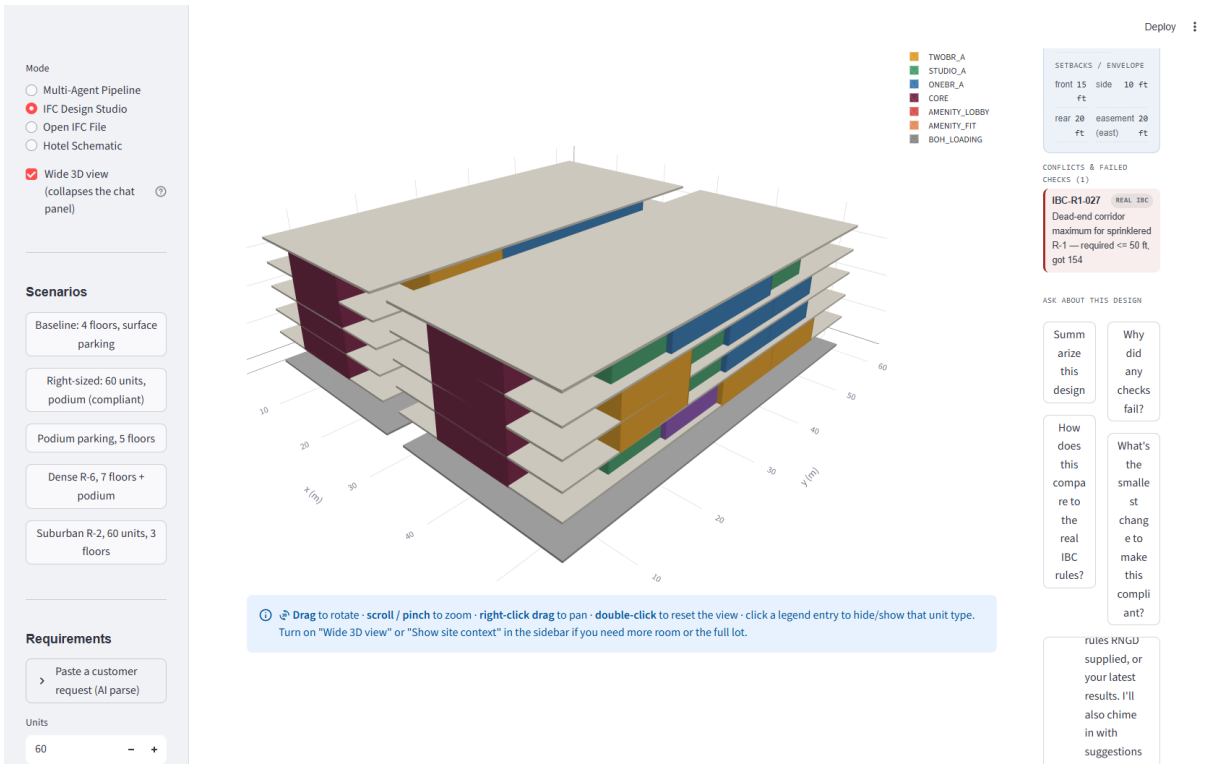


Figure 18: Right-sized scenario in 3D.

7 Open IFC File (a real model)

7.1 What it does

Opens an arbitrary real `.ifc` file, triangulates its geometry, recentres it on the structure (dropping far-away outliers so the building fills the view) and summarises what is in it. The file used is real, supplied for testing.

7.2 Result

Item	Value
File	CDI SOLUTIONS_METAIRIE BANK_R21.ifc (89.4 MB)
Schema	IFC2X3
Project	METAIRIE BANK
Site / building	Default / None
Storeys	FIRST FLOOR (0.0 m), MIDDLE ROOF (14.0 m), HIGH ROOF (16.0 m)
IFC elements	1,537
Meshes triangulated	881
Building diagonal	134.8 m
Outliers excluded from the view	4 (e.g. Night DropBox:Night DropBox:6634177 at 896.5 m)

IFC class	Count
IfcMember	467
IfcBuildingElementProxy	166
IfcPlate	148
IfcWall	144
IfcWallStandardCase	139
IfcFurnishingElement	129
IfcFlowTerminal	128
IfcCurtainWall	36
IfcSlab	25
IfcDoor	23
IfcCovering	22
IfcWindow	7
IfcRoof	2

Notable named items: Wall Sweep (45), Lighting_Fixture-PureEdge_Lighting-Cirrus-D1_Modular_Suspension (36), Lighting_Fixture-PureEdge_Lighting-Cirrus-D1_Modular_Suspension-120_Connector (36), Lighting_Fixture-Focal_Point-Amica_2x4 (19), National - Grin - Guest chair (16), Lighting_Fixture-Focal_Point-Seem_1_Recessed_LED_FSM1 (15), Recessed Lamp - Round - LED (14), Model Text (11), Roller Window Shades (10), CW Cupbd Ohead (9), Knoll_Antenna_Legs_Table_Desks (8), Knoll_Antenna_Tops_Rectangular (8).

7.3 Screenshots

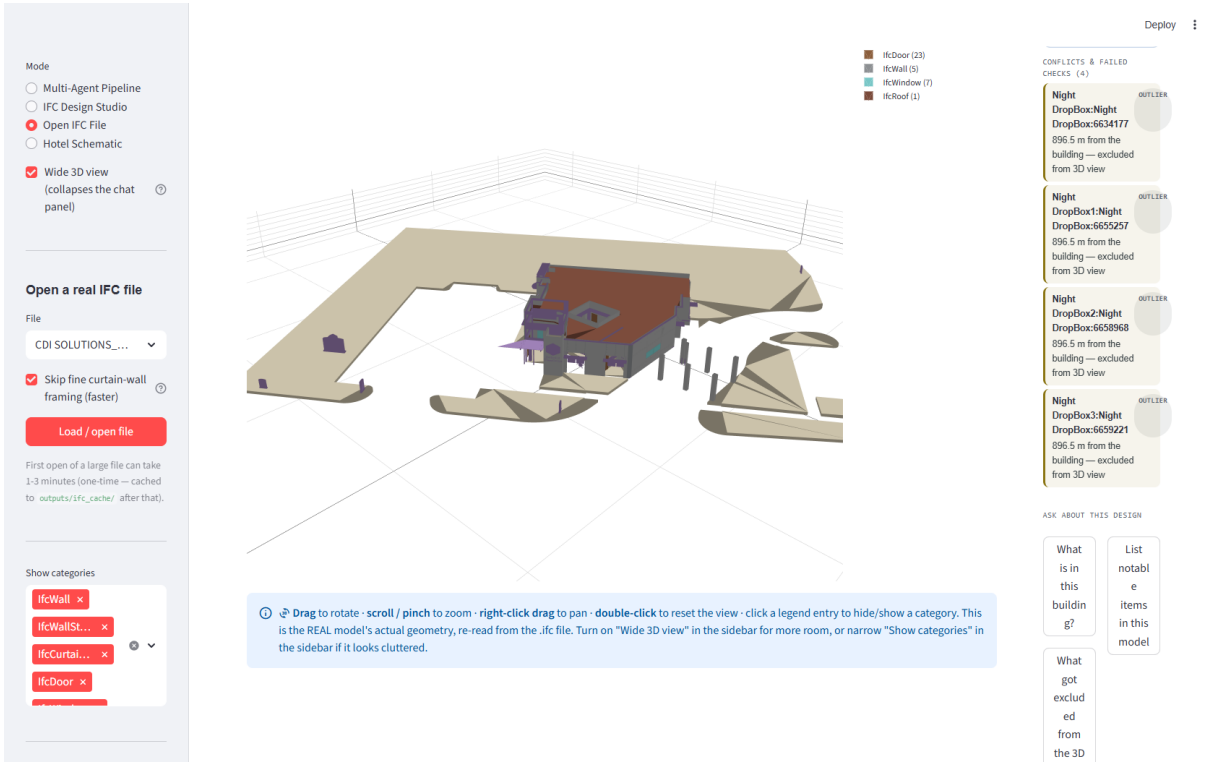


Figure 19: A real IFC file (CDI Solutions, Metairie Bank, IFC2X3, 89 MB) recentred and shown in 3D.

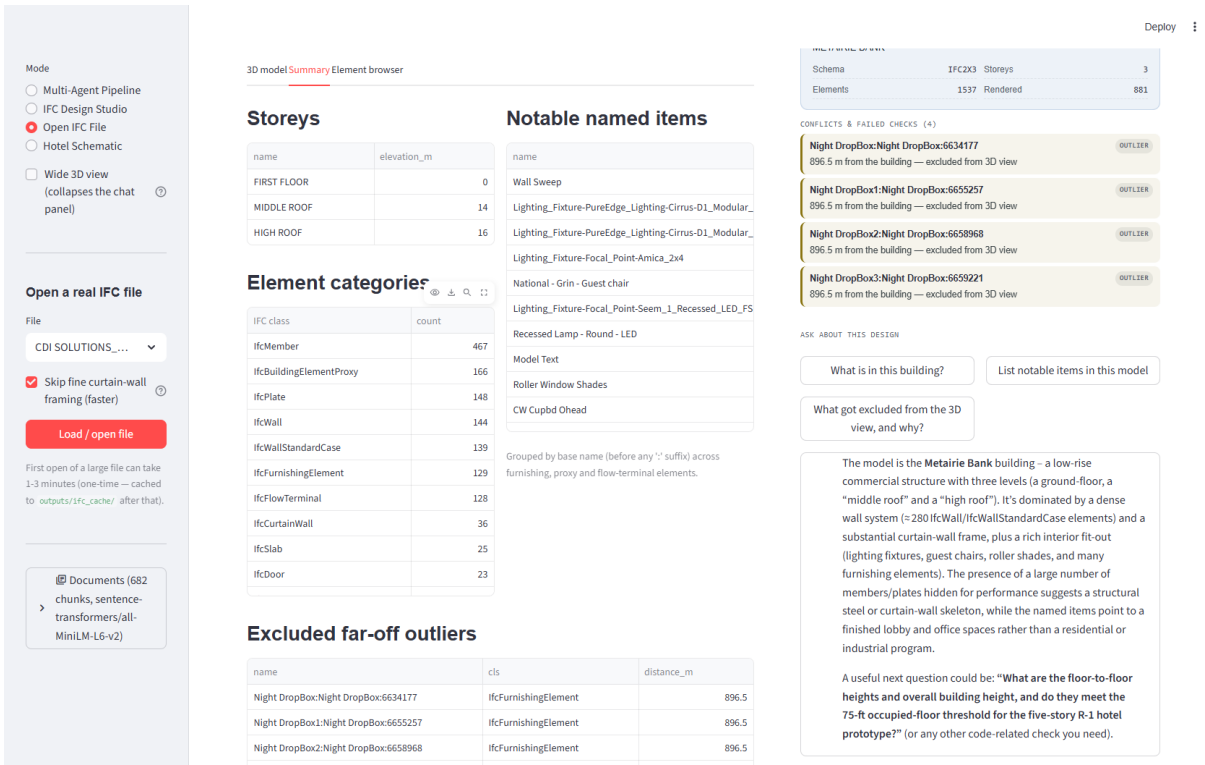


Figure 20: What is in the real model: schema, storeys and element counts by IFC class.

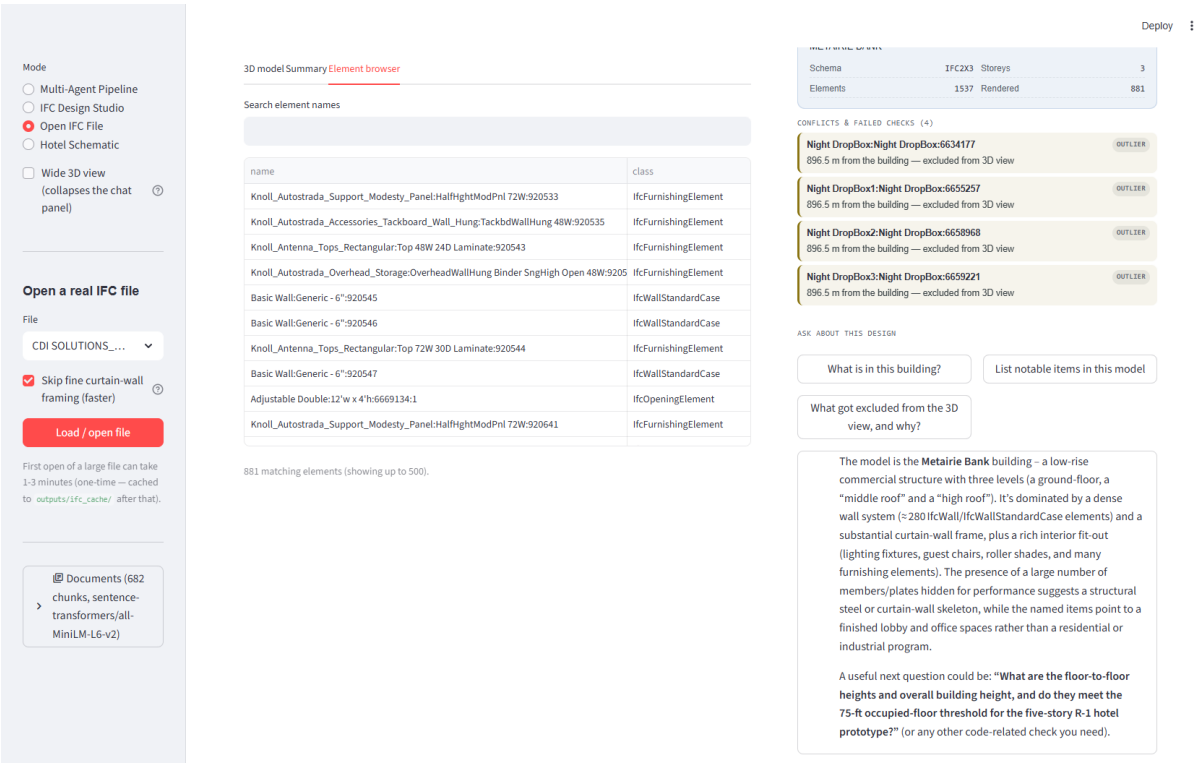


Figure 21: Searchable list of the model's elements.

8 Hotel Schematic (RNGD's first use case)

8.1 What it does

Takes a detailed hotel program — by default the Project Renegade Living + Lodging cover-sheet space allocation (28 rows, 117 keys), or an uploaded / in-app edited CSV or XLSX — and produces a schematic: a Level 1 podium with lobby, BOH and retail, five double-loaded guest floors with a core and a stair at each end, a code screen against RNGD's real IBC rule CSV and the 2010 ADA guest-room tables, and an 18 x 12 in drawing set in RNGD's L+L sheet format.

8.2 Headline results

Keys	117 on 5 guest floors
Net program area	59,350 SF (28 rows)
Gross area: source sheets / meeting rule / drawn	74,195 / 85,300 / 76,508 SF
Massing	podium 234 x 58 ft, tower 217 x 58 ft, 69.75 ft to roof
Code screen	13 pass, 1 fail, 1 review, 1 assumed
Drawing set	10 sheets, 18 pages, generated 2026-10-05 16:01:06

8.3 Code and ADA screen

Rule	Check	Required	Provided	Status
-	Keys placed	117	117	PASS
IBC-R1-007	Stories above grade	≤ 5 (Type IIA, sprinklered R-1)	6	FAIL
IBC-R1-005	Building height (ft)	≤ 85 (Type IIA)	69.8	PASS
IBC-R1-010	Area per story (SF)	≤ 72,000 (Type IIA)	13,575	PASS
IBC-R1-012	High-rise screen (ft to top floor)	≤ 75	59.0	PASS
IBC-R1-015	Exits per story	2 for 1-500 occupants	2 stairs; max load 316	PASS
IBC-R1-018	Exit separation (ft)	≥ 1/3 diagonal = 75	207	PASS
IBC-R1-023	Exit access travel (ft)	≤ 250 (sprinklered)	130	PASS
IBC-R1-025	Corridor width (in)	≥ 44	72	PASS
IBC-R1-027	Dead-end corridor (ft)	≤ 50 (sprinklered R-1)	5	PASS
IBC-R1-028	Egress door clear width (in)	≥ 32	34 (3’-0” leaf)	PASS
IBC-R1-003	Automatic sprinklers	NFPA 13 required	assumed	ASSUMED
ADA-224.2	Accessible guest rooms (mobility)	≥ 7 for 117 rooms (5 + 2 with roll-in)	7	PASS
ADA-224.4	Rooms with communication features	≥ 12	not designated	REVIEW
ADA-224.5	Dispersion across room types	spread over 3 types	3 types: king, suite, two-bed	PASS
-	Fits buildable envelope	170 x 265 ft	234 x 58 ft	PASS

Notes

- **IBC-R1-007**: Level 1 counts as a story above grade plane. A Type IA podium (IBC 510.2) could allow 5 stories above it.
- **IBC-R1-023**: Half the stair-to-stair run plus the room depth.
- **ADA-224.2**: Confirm which 2 of the ADA units get roll-in showers (Table 224.2).
- **ADA-224.4**: Designate 12 rooms; at most 10% of the mobility rooms may count (224.5).

Suggested next steps for the open items

- **Stories above grade**: Use a Type IA podium for Level 1 (IBC 510.2) so the five guest floors count as a separate building, or drop to four guest floors and lengthen the wing.
- **Rooms with communication features**: Mark the communication-feature rooms in the program (visual alarms, notification devices).

8.4 Levels

Level	Floor at (ft)	Footprint (ft)	Gross drawn (SF)	Gross source (SF)	Keys
Level 1	0.00	234 x 58	13,575	12,485	0
Level 2	16.00	217 x 58	12,587	12,342	24
Level 3	26.75	217 x 58	12,587	12,342	24
Level 4	37.50	217 x 58	12,587	12,342	23
Level 5	48.25	217 x 58	12,587	12,342	23
Level 6	59.00	217 x 58	12,587	12,342	23

8.5 Detailed program

Group	Type	Count	Each SF	Total SF	Size	Beds	Sleeps	Placed on	Occ.
Shared	Corridor	1	1,335	1,335				L1, L2, L3, L4, L5, L6	
Shared	Gym	1	968	968				L1	20
Shared	Lobby	1	2,865	2,865				L1	191
Shared	Mens	1	205	205				L1	
Shared	Womens	1	200	200				L1	
BOH	BOH	1	69	69				L1	1
BOH	Hotel Kitchen / Pantry	1	331	331				L1	2
BOH	Housekeeping	1	103	103				L1	1
BOH	Ice	5	59	297				every guest floor	1
BOH	IT	5	74	372				every guest floor	2
BOH	Jan	5	54	269				every guest floor	1
BOH	Janitor	1	107	107				L1	1
BOH	Loading	1	282	282				L1	1
BOH	MEP Service	1	309	309				L1	2
BOH	Stair	12	156	1,868				2 per level (stacked)	
BOH	Supplies	5	79	394				every guest floor	2
Tenant	White Box Retail	1	3,533	3,533				L1	59
Hotel Units	ADA Corner Suite	1	852	852	32.8’ x 26’	1 king + sofa	4	L4	5
Hotel Units	ADA King Spa	2	460	920	17.7’ x 26’	1 king	4	L5, L6	5
Hotel Units	ADA QQ	2	446	892	17.2’ x 26’	2 queens	8	L5, L6	5
Hotel Units	ADA Suite	2	639	1,278	24.6’ x 26’	1 king + sofa	8	L3, L4	7
Hotel Units	Corner Suite	2	864	1,728	33.2’ x 26’	1 king + sofa	8	L2, L3	9
Hotel Units	King	69	343	23,692	13.2’ x 26’	1 king	138	every guest floor	119
Hotel Units	King Spa	3	460	1,379	17.7’ x 26’	1 king	6	L2, L3, L4	7
Hotel Units	QQ	33	344	11,342	13.2’ x 26’	2 queens	132	every guest floor	57
Hotel Units	Suite	3	640	1,920	24.6’ x 26’	1 king + sofa	12	L2, L5, L6	10
Business	BOH / Bus	1	1,718	1,718				L1	12
Business	Luggage	1	122	122				L1	1

8.6 Room mix by floor

Type	Level 2	Level 3	Level 4	Level 5	Level 6	Total
Corner Suite	1	1	0	0	0	2
King Spa	1	1	1	0	0	3
QQ	7	7	7	6	6	33
King	14	14	13	14	14	69

Type	Level 2	Level 3	Level 4	Level 5	Level 6	Total
Suite	1	0	0	1	1	3
ADA Suite	0	1	1	0	0	2
ADA Corner Suite	0	0	1	0	0	1
ADA QQ	0	0	0	1	1	2
ADA King Spa	0	0	0	1	1	2
Keys per floor	24	24	23	23	23	117

8.7 App screenshots

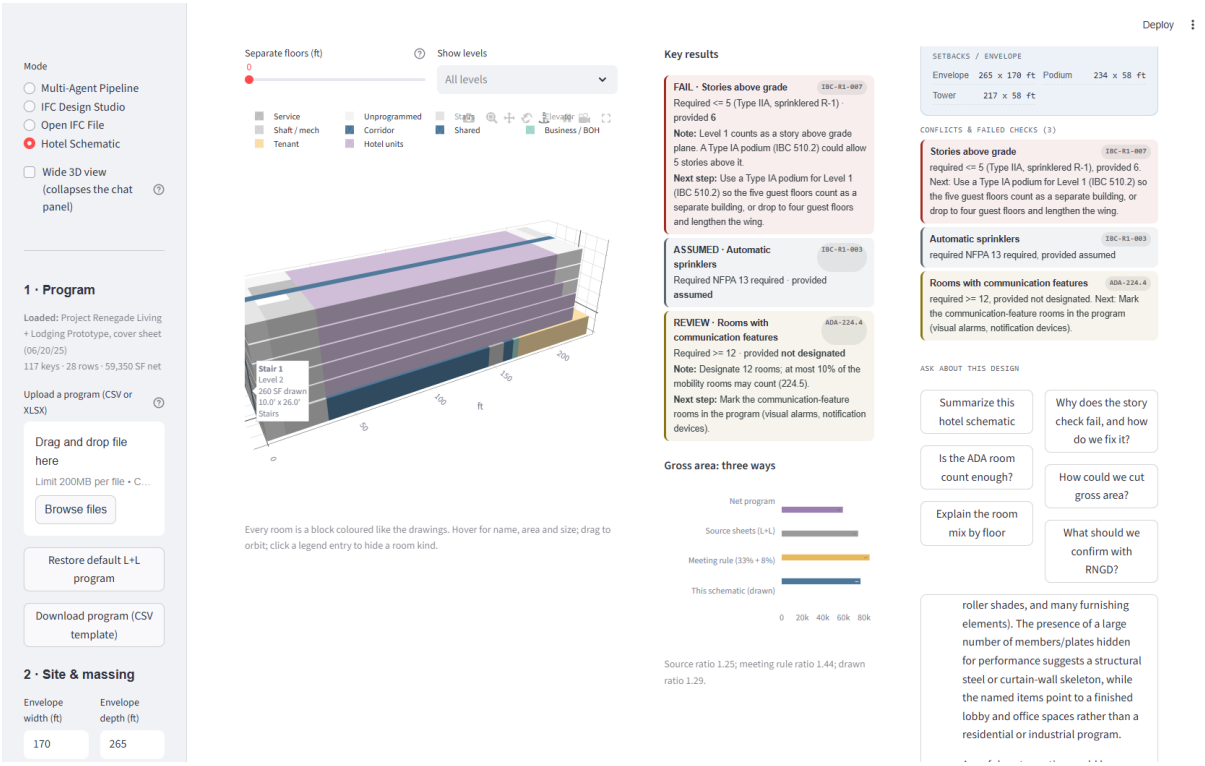


Figure 22: Hotel Schematic workspace: KPI tiles, the room-by-room 3D model and the key results with suggested fixes.

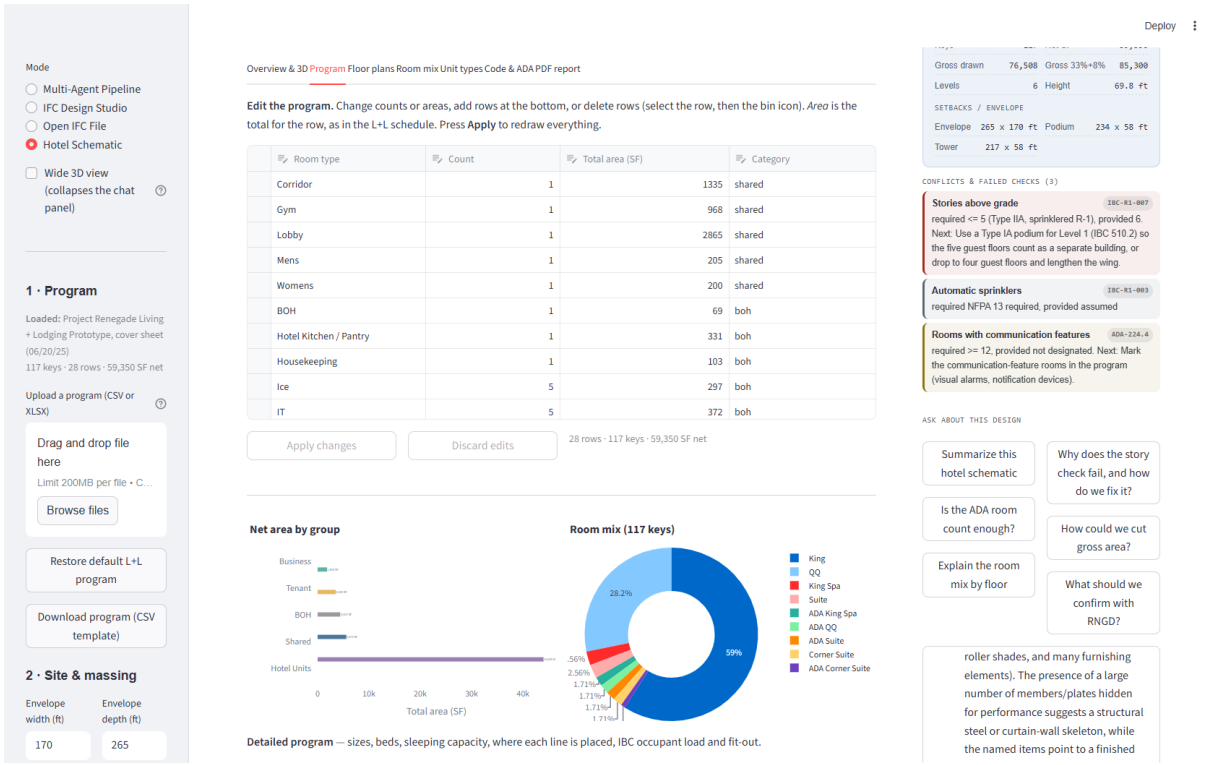


Figure 23: The editable program (Project Renegade L+L space allocation, 28 rows) with area and room-mix charts.

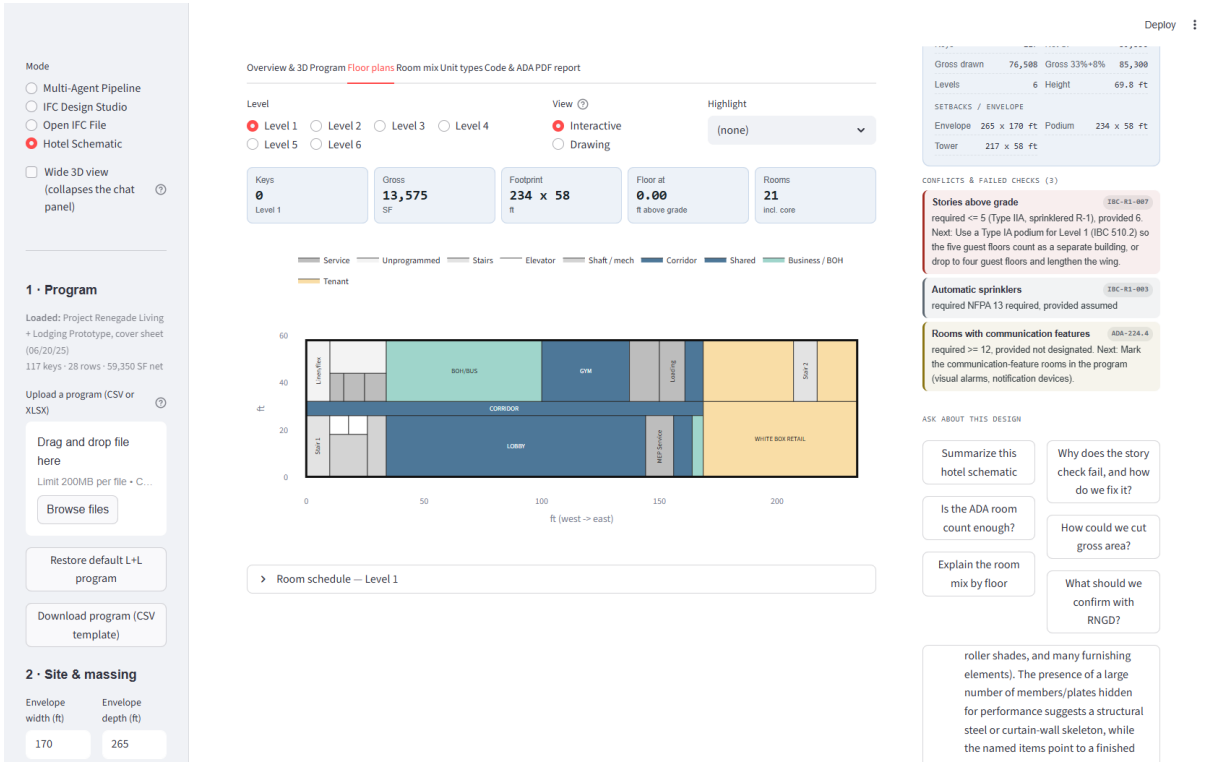


Figure 24: Interactive floor plan of Level 1 (hover shows each room's area and size).

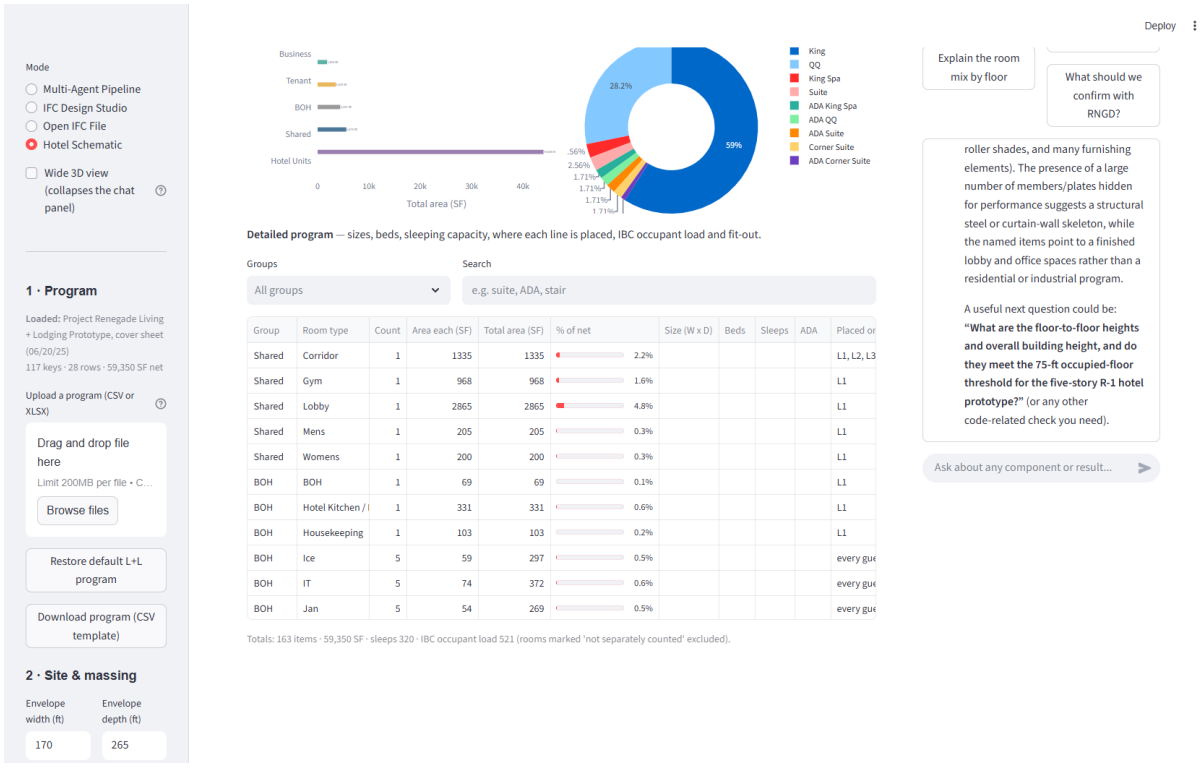


Figure 25: Detailed program: per-room size, beds, sleeping capacity, placement, IBC occupant load and fit-out.

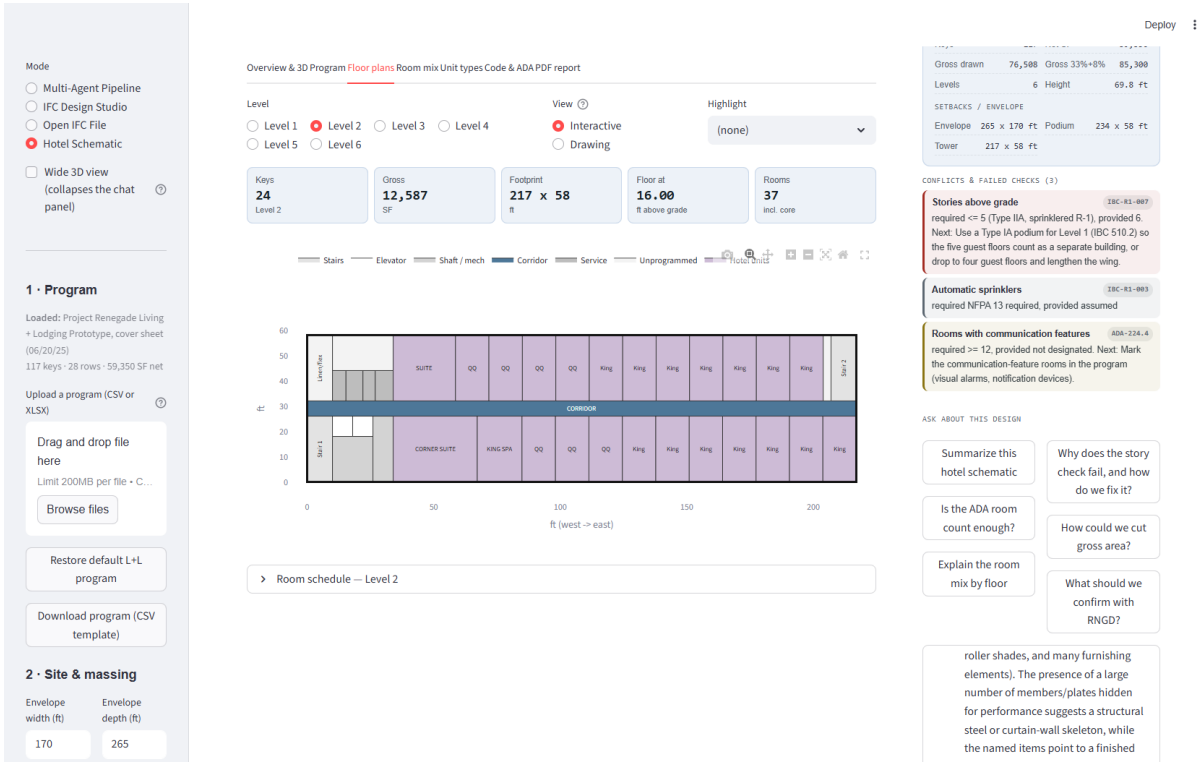


Figure 26: Interactive plan of Level 2, a typical guest floor (24 keys).

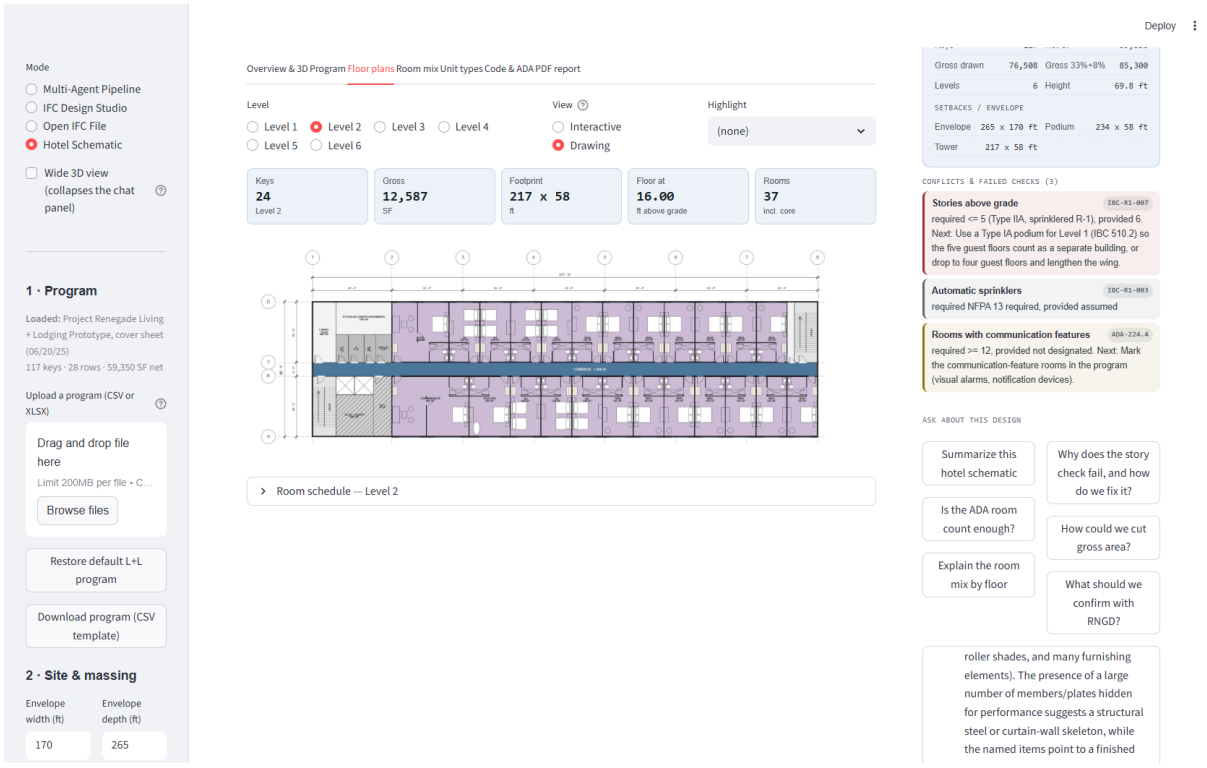


Figure 27: The same level in drawing style (as used on the PDF sheets).

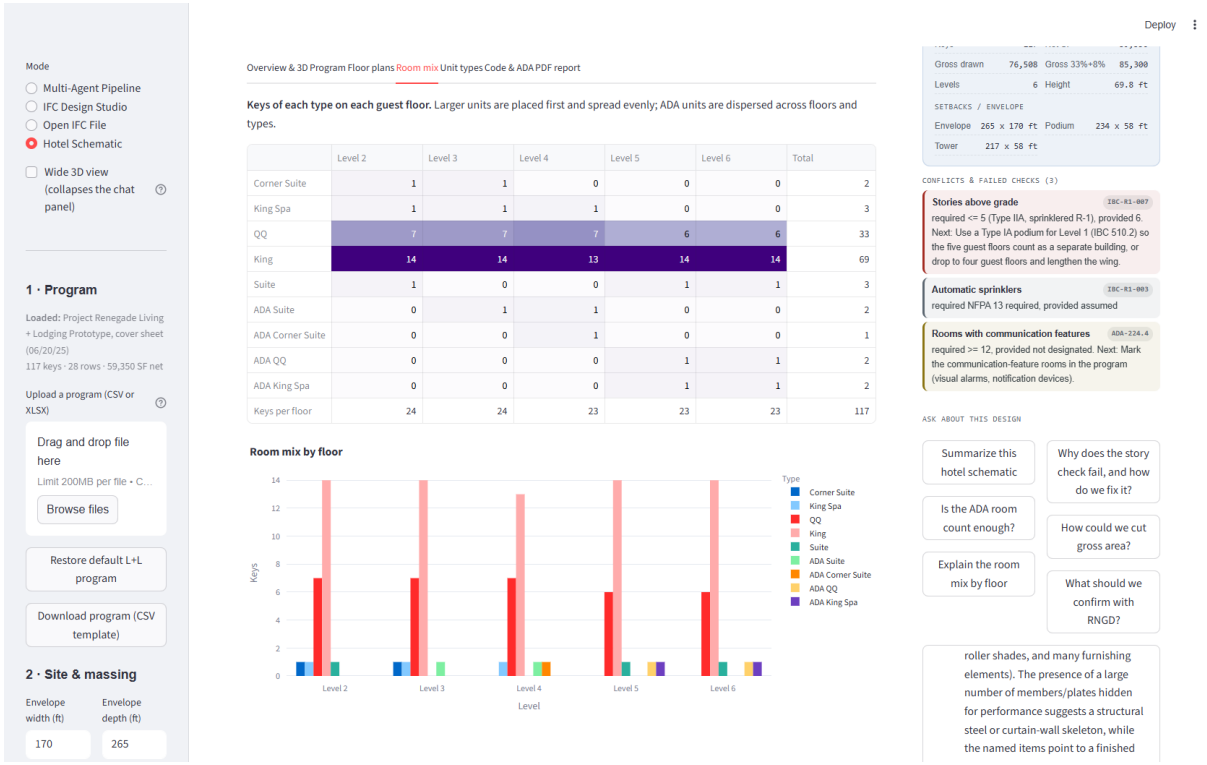


Figure 28: Keys of each room type on each guest floor.

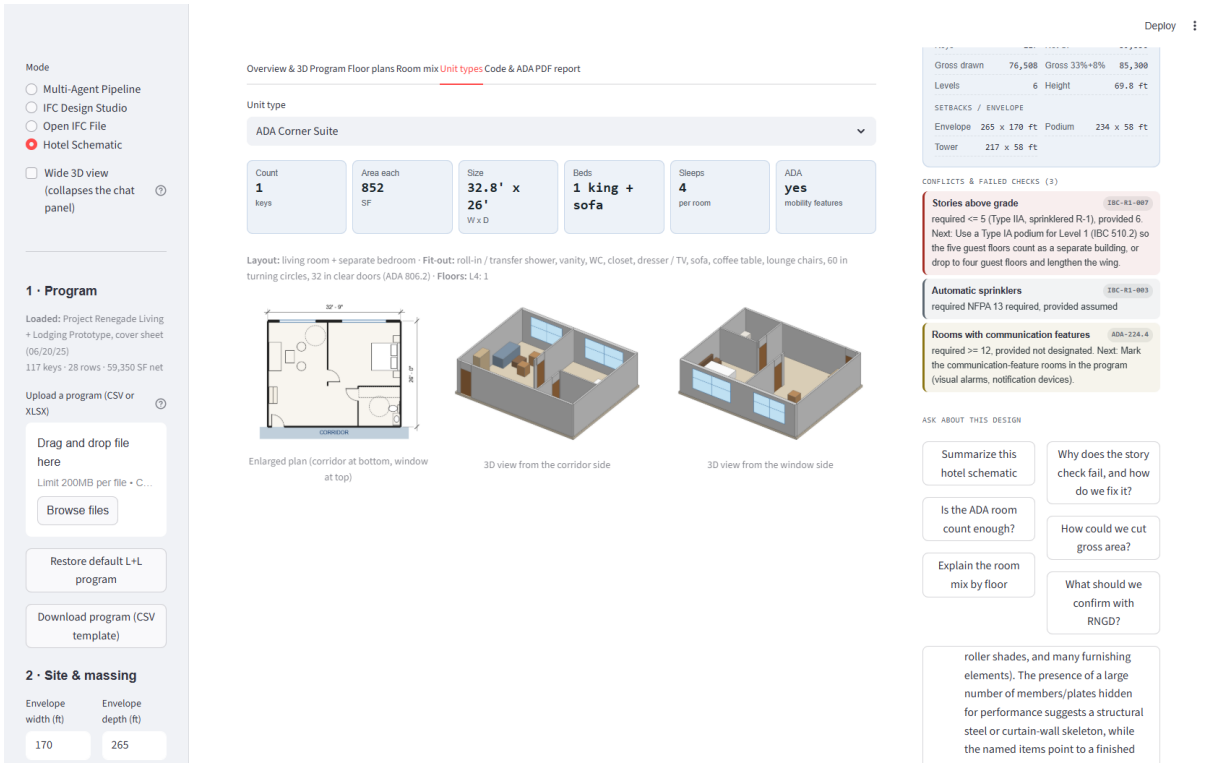


Figure 29: Unit type detail: enlarged plan and two 3D interior views.

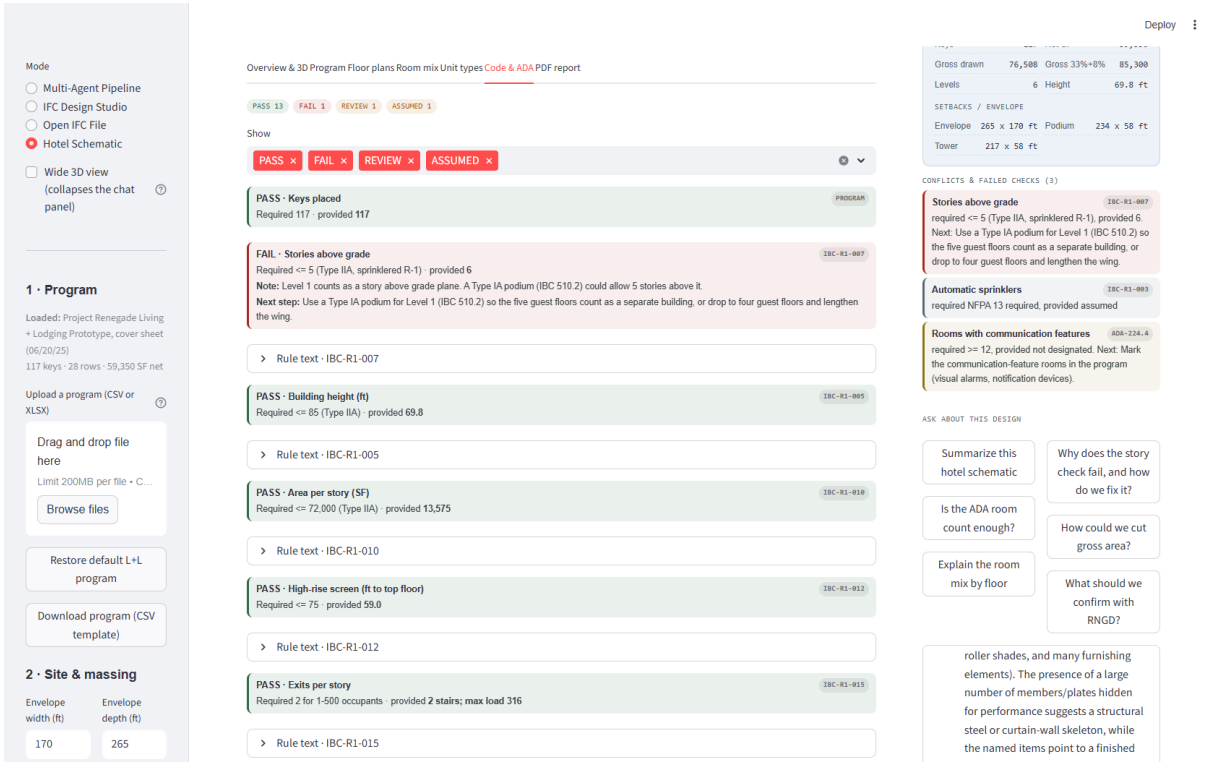


Figure 30: Code screen against RNGD's IBC rule set and the 2010 ADA guest-room tables, with rule text.

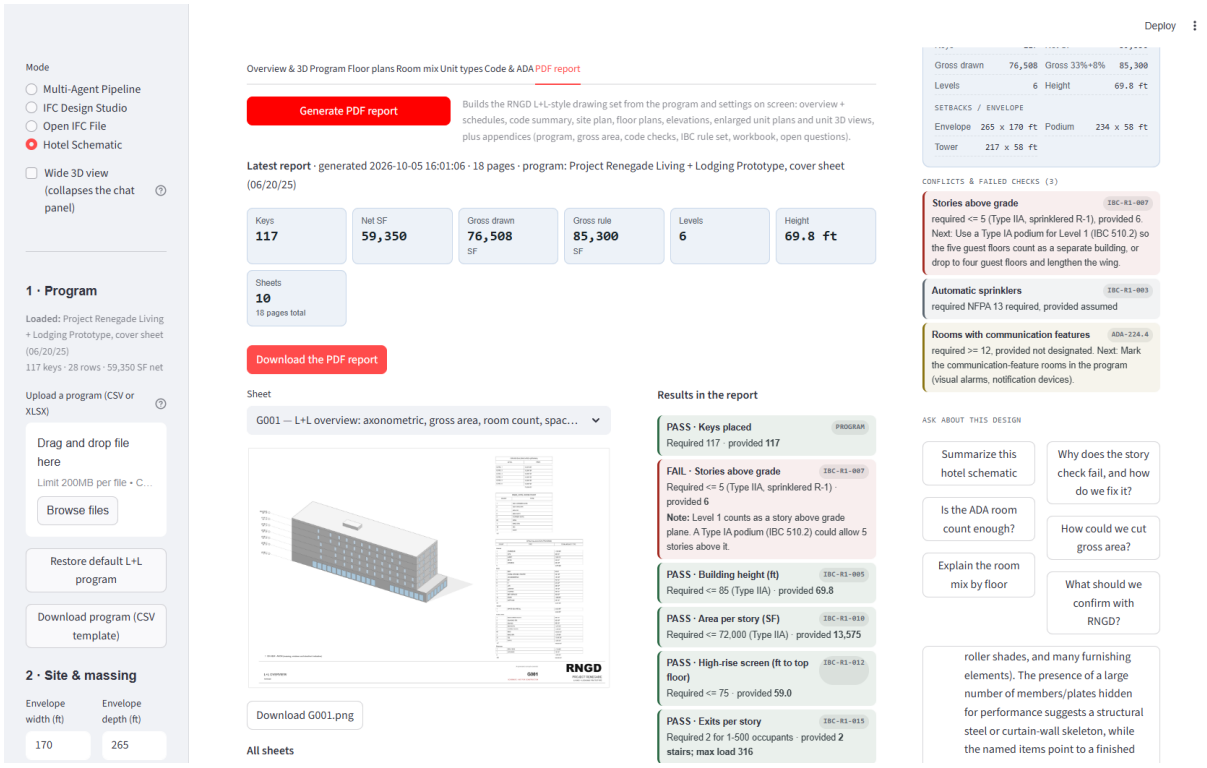


Figure 31: PDF report tab after generating: summary numbers and the results that went into the report.

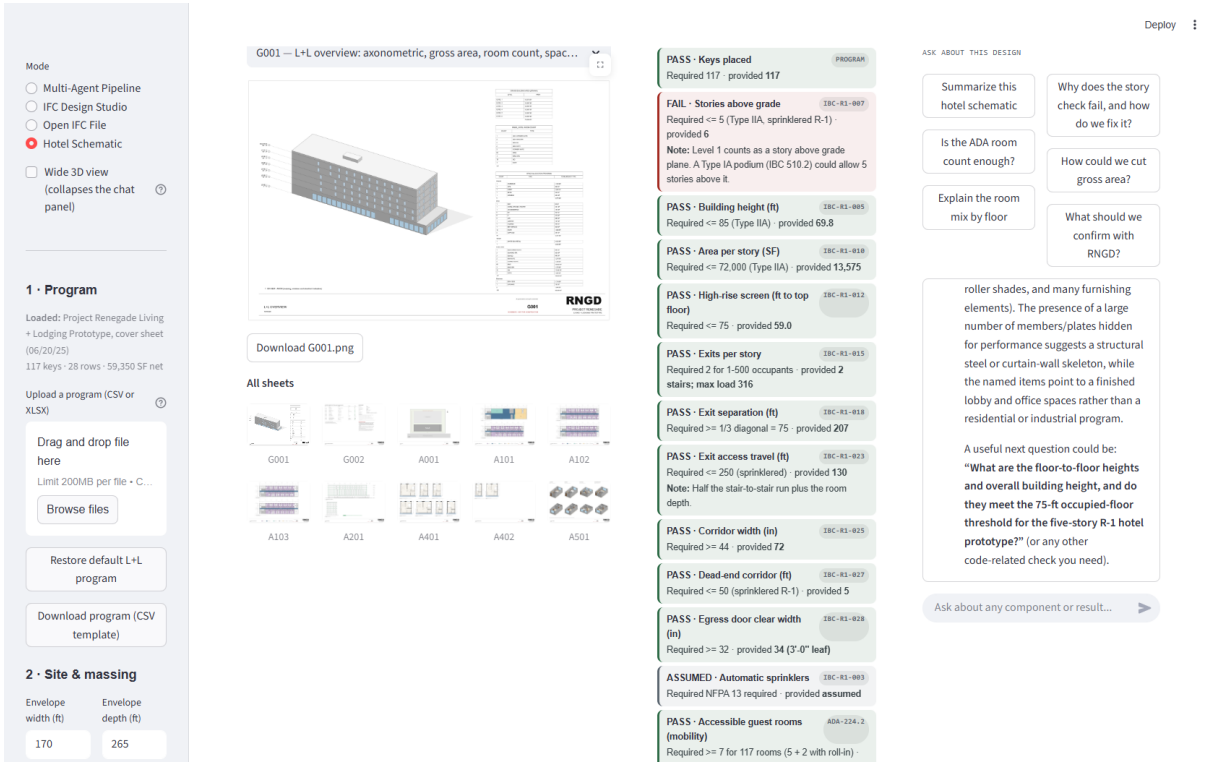


Figure 32: Sheet gallery of the generated drawing set.

8.8 Generated drawing set

The PDF report (`reports/final/RNGD_final_schematic_report.pdf`) contains these sheets:

Sheet	Content
G001	L+L overview: axonometric, gross area, room count, space allocation
G002	Code screen and program summary
A001	Site plan
A101	Floor plans: Levels 1 & 2
A102	Floor plans: Levels 3 & 4
A103	Floor plans: Levels 5 & 6
A201	Elevations with indicative panel breakdown
A401	Enlarged unit plans
A402	Enlarged unit plans
A501	Unit 3D views (interiors)

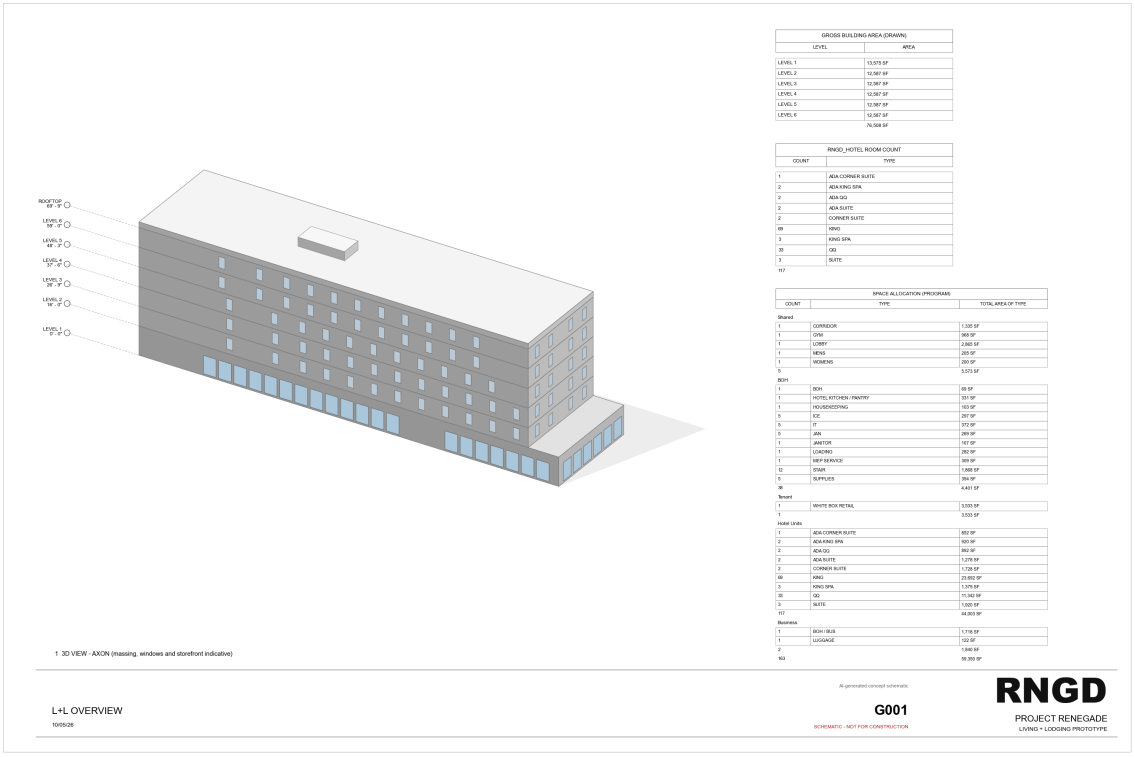


Figure 33: G001 - L+L overview: axonometric, gross area, room count, space allocation (from the generated PDF drawing set)

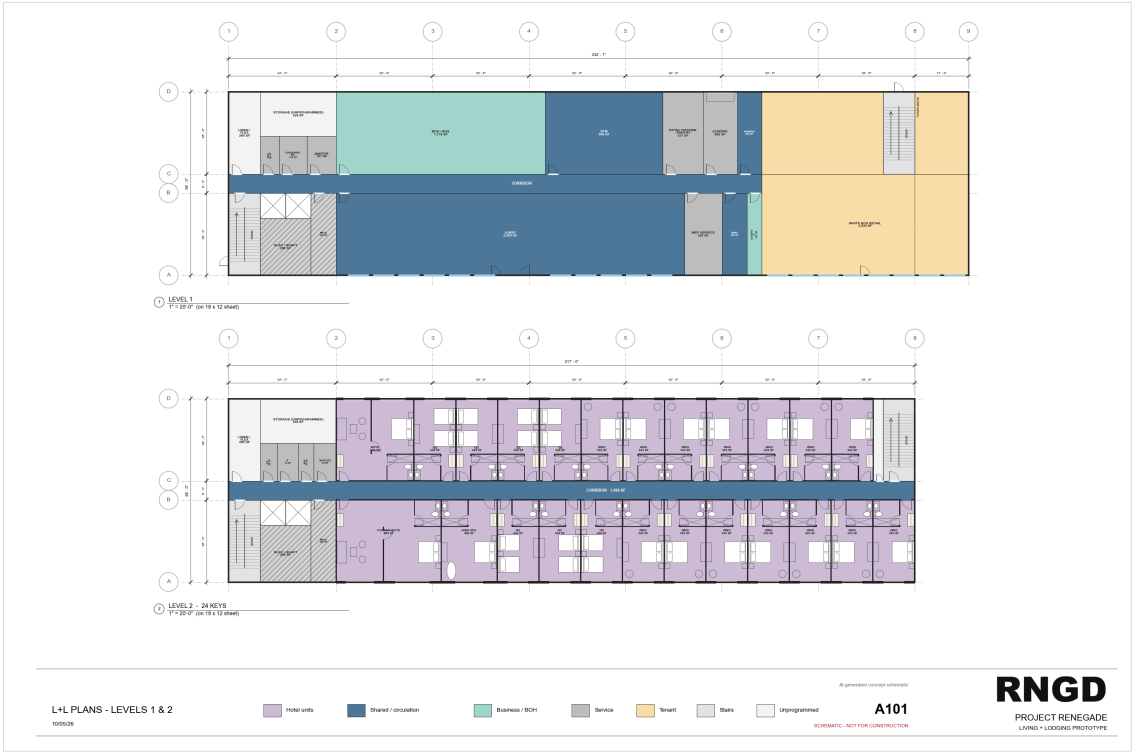


Figure 34: A101 - Floor plans: Levels 1 & 2 (from the generated PDF drawing set)



Figure 35: A102 - Floor plans: Levels 3 & 4 (from the generated PDF drawing set)

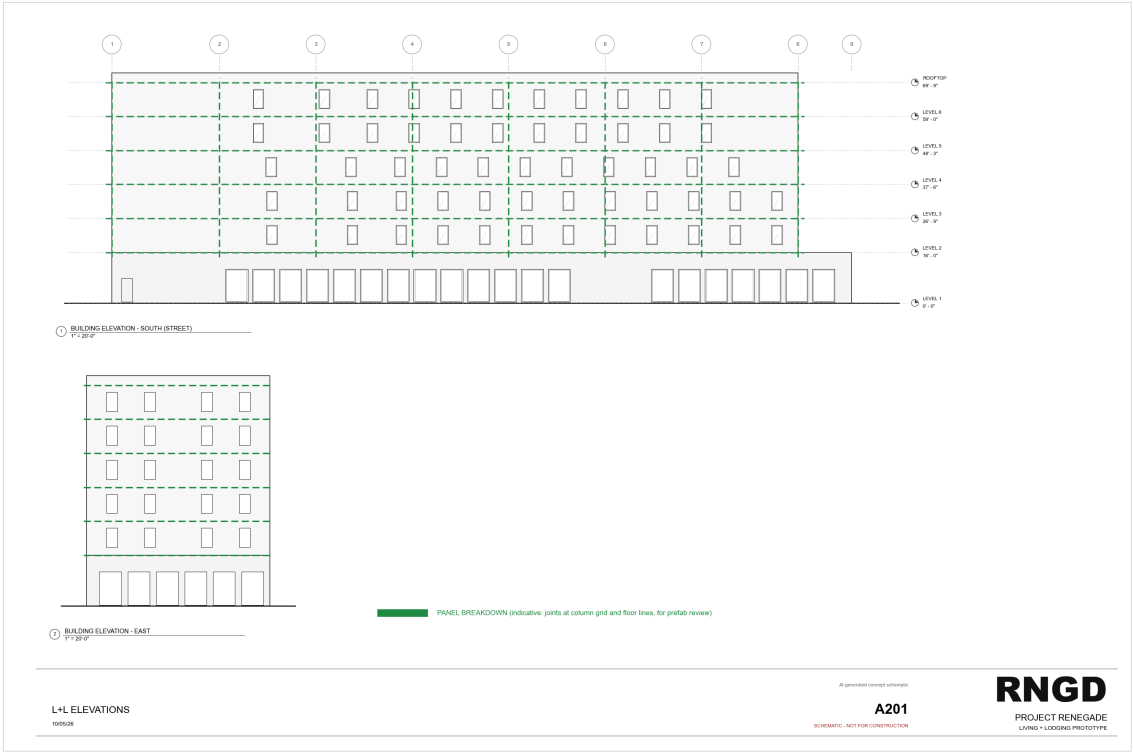


Figure 36: A201 - Elevations with indicative panel breakdown (from the generated PDF drawing set)



Figure 37: A401 - Enlarged unit plans (from the generated PDF drawing set)



Figure 38: A501 - Unit 3D views (interiors) (from the generated PDF drawing set)

9 Layout optimization: from the reference GA to the updated optimizer

9.1 Summary

RNGD's design-optimization prototype (Büşra's genetic algorithm, files in `QA/`) takes a multi-level 1 ft site grid and a room program and evolves a corridor and a room placement for 500 generations. This project rebuilt that optimizer in four steps and measured every step on the same three sites with the same program and the same evaluator (Table 22):

- **A, the reference GA** as delivered: 171 rooms placed over the three sites, 24 of the guest rooms without a window.
- **C, the updated GA**: a memetic GA on a new architectural decoder, no LLM in the fitness: 197 rooms, 0 without a window, mean fitness 0.928 against 0.863.
- **D, systematic search** (the default now): the same decoder, driven by enumeration and pattern search instead of crossover: 192 rooms, 0 without a window, mean fitness 0.922, in 59 s for all three sites and with the same answer on every run.

Per site (A $\rightarrow$ D): fp1: 61 $\rightarrow$ 72 rooms, 7 $\rightarrow$ 0 windowless, fitness 0.880 $\rightarrow$ 0.950; fp2: 53 $\rightarrow$ 60 rooms, 8 $\rightarrow$ 0 windowless, fitness 0.860 $\rightarrow$ 0.904; fp3: 57 $\rightarrow$ 60 rooms, 9 $\rightarrow$ 0 windowless, fitness 0.851 $\rightarrow$ 0.911. Variant B, the first attempt, kept the reference idea (a GA that moves free corridor segments and packs rooms) and placed 174 rooms in total with 14 windowless guest rooms in 336 s: better scoring alone did not fix the plans, which is why the representation was changed.

The same engine now also runs inside **Hotel Schematic**, where the shape of the building, the position of the core and stairs and the number of floors are generated instead of given (Section 9.6).

9.2 Architecture

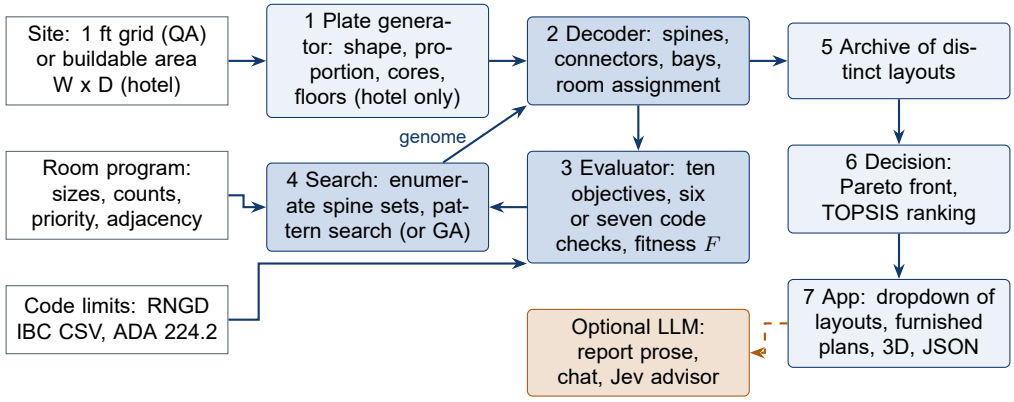


Figure 39: Architecture of the layout optimizer. The darker boxes form the optimization loop and are deterministic. The language model (orange) sits outside the loop: it only writes report prose, answers chat questions, or, if a TypeSafe key is configured, advises on the final shortlist.

#	Component	File	What it does
1	Plate generator	`optimize/typology.py`	Turns the hotel program into rooms with BaaP sizes; generates Bar, L, U, T, H and Courtyard plates inside the buildable area, with the elevator core and stairs placed by scheme (ends, centre, knuckle); one site grid per plate.
2	Decoder	`optimize/planner.py`	Genome $\rightarrow$ corridor spines $\rightarrow$ orthogonal connectors $\rightarrow$ modular bays $\rightarrow$ rooms assigned to bays. Every output is a valid plan.
3	Evaluator	`optimize/metrics.py`	Ten objectives in $[0, 1]$, the code checks (exits, exit separation, travel, corridor width, dead end, ADA; stories for the hotel) and the fitness F. Limits come from RNGD's IBC CSV.
4	Search	`optimize/search.py`, `explore.py`	Systematic enumeration of spine sets plus pattern search (default) or the memetic GA; for the hotel, all plates are screened in parallel processes and the best of each shape refined.
5	Archive	`optimize/explore.py`	Keeps every distinct layout with its objectives and criteria.
6	Decision	`optimize/decide.py`	Pareto front and TOPSIS ranking over twelve criteria; optional Jev advisor (TypeSafe) on the shortlist only.
7	Views	`optimize/plan_view.py`, `hotel_opt_ui.py`, `optimizer_ui.py`	Dropdown of the top or all layouts, furnished interactive plans, 3D massing, comparison charts, JSON export.
–	Reference loader	`optimize/baseline.py`	Rebuilds the reference designs from `QA/floorplan*.json` so they are scored by the same evaluator (their shared-wall lengths reproduce the values stored in the files).
–	Report writer	`optimize/report.py`	One LLM call (`anthropic/claude-sonnet-5.5` through OpenRouter) writes the prose of the per-run PDF from the computed data and retrieved code passages; every number in its text is traced back to the data.

Only the last row uses a language model. Steps 1 to 6 are plain Python on NumPy, SciPy and scikit-image arrays, so the same inputs always give the same layouts and the same ranking.

9.3 What was wrong, and how it was solved

The problems below were seen in the reference results, in the first version of this app's optimizer (which followed the same idea and added an LLM reviewer to the fitness) and in the team review of 10/08/2026.

Observed problem	Cause	What was changed	Evidence in this run
Guest rooms with no window; corridors loaded on one side only (review in the 10/08 meeting).	The corridor genome places segments anywhere; rooms are packed wherever they touch a corridor, so the distance to the facade is left to chance.	Decoder: corridor spines are chosen from lines scored for double-loaded window rooms (score $s(k)$); guest rooms may only take bays with a window (hard rule).	Windowless guest rooms over the three sites: 24 (A) $\rightarrow$ 0 (D).
Floor area left unused; rooms rotated or isolated.	Rooms are placed one by one as free rectangles, so leftovers between rooms cannot be used.	Modular bays of one room frontage M along the whole corridor; rooms take whole bays, the last bay absorbs the remainder; unused bays are reported as flex space.	Unused share and corridor share per variant in Table 22.
Not clear whether the requested keys can fit at all.	No bound was computed; the GA simply reports what it placed.	Area and window upper bounds, shown next to every result.	Table 24.
Crossover produces plans that do not make sense; slow convergence.	Half of one plan spliced onto half of another: the value of a corridor depends on the other corridors.	Search over a small structured space: enumerate spine sets, then first-improvement pattern search. The GA is kept as an option and now runs on the same decoder.	At the same budget D is within 0.010 fitness of the GA (C) with no crossover and no population: 192 against 197 rooms. Most of the gain over A comes from the decoder, not from the search.
With crossing corridors, facade strips stayed empty (seen on the square site fp2 in the first version of the new decoder: 39 rooms).	Bays were built corridor by corridor, so the windowless interior side of the first corridor took the strip that the next corridor's window rooms needed.	Two passes over all corridors: window bays first, windowless bays in what is left.	fp2: 60 rooms (D) and 62 (C), all guest rooms with a window.
Corridor stubs beyond the last room; dead ends over the 50 ft limit.	Spines were cut to the bays they could serve, not to the rooms actually placed.	After the assignment each spine is cut back to the doors and junctions it serves; the dead end is measured to the nearest exit door or through-route.	Longest dead end, D: 46 ft; A: 41 ft (same measure).
Runs take minutes; results change between runs.	500 generations in the reference; an LLM reviewer inside the fitness in the earlier version of this app (about 10 minutes, cloud calls).	No LLM in the fitness; deterministic seeds; cached static scores in the assignment; plates evaluated in parallel processes.	Variant D: 59 s for the three sites; identical result on a repeat run: yes.
Egress and accessibility not part of the score.	The reference score rewards placed rooms and contiguity.	Ten objectives including exit travel, dead ends, ADA Table 224.2 count and dispersion, core access; limits read from RNGD's IBC CSV.	All variants pass the checks on these sites; the checks are now reported for every layout.
The shape, the core and the floor count are inputs.	The grid fixes the L-shaped plate and both cores.	Hotel Schematic: plates (six shapes), core schemes (three) and the floor count are generated and searched; a dropdown lists the top or all layouts.	740 distinct layouts in 36 s (Section 9.6).

9.4 The mathematics

Problem. A site is a stack of L levels of 1 ft cells. On each level a set Ω of cells is *placeable*; cores (stairs, elevators) and structure are fixed in the QA sites and generated for the hotel. The program is a list of room instances $i = 1..N$, each with a type $t(i)$, nominal frontage f_i and depth d_i (area $\bar{a}_i = f_i d_i$), a priority p_i, a set of allowed levels and flags (needs a window, wants a corner, wants to be near a core). A layout chooses a corridor $C \subset \Omega$ (the same on every level, so floors stack) and an axis-aligned rectangle for some of the rooms, such that rectangles do not overlap each other or C, lie inside Ω, and each room has a door on C. The goal is to maximise the fitness F below.

Decoder: from a short genome to a full layout. The search never moves rooms directly. It works on a genome $g \in [0,1]^{44+N}$ (ten candidate spines with four genes each: two trims, a shift, an on/off gene; four connector genes; one order key per room) and the decoder builds the layout:

1. **Module.** The bay width is the weighted median guest-room frontage, $M = \text{median}_w\{f_t\}$ with weights equal to the room counts; D is the median guest-room depth and w the corridor width.
2. **Candidate spines.** Every row and column k that can hold a straight corridor band gets the score

$$s(k) = \sum_{x \in \text{band}(k)} \big[\, g_a(x) + g_b(x) + 1.5\, n_a(x) + 1.5\, n_b(x) + 0.5\, r_a(x) + 0.5\, r_b(x) \,\big],$$

   where, for the two sides a and b of the band at position x, $g = 1$ if a room at least $0.92\, d_{\text{min}}$ deep fits, $n = 1$ if in addition the facade is about one room depth away (a window room), and $r = 1$ if it is within the deepest room's depth. So a line that is double-loaded with window rooms on both sides scores $2 \times (1 + 1.5 + 0.5) = 6$ per foot, a single-loaded line 3, and a line through deep space with no facade 2. The best non-overlapping lines (at least $d_{\text{min}} + w$ apart) become the ten candidates.
3. **Connectors.** Active spines are trimmed to their last room and joined to each other and to every core with orthogonal L-shaped paths (never diagonal); pieces that stay unconnected are dropped.
4. **Bays.** Each side of each straight corridor run of length ℓ is cut into $n = \lfloor \ell / M \rfloor$ bays. The remainder $\rho = \ell - nM$ is absorbed by the last bay if $\rho < 0.4M$ and becomes a narrow bay otherwise. A bay has a window if its back wall is on the facade. Bays are built in two passes over all corridors, window bays first and windowless bays in what is left, so the interior side of one corridor never takes the facade strip that another corridor's window rooms need.
5. **Assignment.** Rooms are taken in the order of $0.06\, p_i + g_i$ (priority, then the order key) and each takes

$$k_i = \max\Big(\text{round}\,\frac{\bar{a}_i}{MD},\ \Big\lceil \frac{0.85 f_i}{M} \Big\rceil\Big)$$

   consecutive free bays on one allowed level, the run with the highest score

$$\sigma = W_{\text{ext}}\,[\text{window}] - W_{\text{nw}}\,[\text{no window}] + W_{\text{corner}}\,[\text{corner}] + W_{\text{core}}\Big(1 - \tfrac{\min(\delta,150)}{150}\Big) + W_{\text{adj}}\,A$$
$$+ W_{\text{stack}}\,\tfrac{\min(S,2)}{2} + W_{\text{grp}}\,[\text{same type next door}] - 1.2\,\max(0, o - 1.2),$$

   with δ the corridor distance from the door to the nearest core, A the adjacency credit, S the number of floors with the same room type in the same bay, and o the ratio of the bay area to the room's nominal area. Two hard rules apply: a guest room is only placed where every bay has a window, and only if the run gives at least 85% of its nominal frontage and depth.

After the assignment, each spine is cut back to what it serves (the doors of placed rooms and its junctions with other corridors and cores), which removes corridor stubs and shortens dead ends. The dead end E is the longest corridor distance from any corridor cell to the nearest exit door or through-route between two exits.

Every decoded layout is therefore valid by construction (straight corridors, rooms perpendicular to them, no overlaps, all rooms on the corridor), and the search only has to decide *which* corridors exist and *in what order* rooms claim bays.

Fitness. Ten objectives $f_k \in [0,1]$ (1 is best) are combined with weights w_k:

$$F = \frac{\sum_k w_k f_k}{\sum_k w_k} \;-\; \tfrac{1}{2}\max\big(0,\ \min(n_c, 2) - c\big),$$

where n_c is the number of cores and c the number the corridor reaches (an exit that is not connected costs half the scale). With ρ the corridor share of the plate, T the longest exit travel and E the longest dead end:

$$f_{\text{place}} = \frac{\sum_{i\ \text{placed}} p_i \min(a_i, \bar{a}_i)}{\sum_i p_i \bar{a}_i}, \qquad f_{\text{day}} = \frac{\#\{\text{rooms needing a window that have one}\}}{\#\{\text{rooms needing a window}\}},$$

$$f_{\text{eff}} = \text{clip}\Big(1 - \frac{\rho}{0.30}\Big), \qquad f_{\text{egress}} = 1 - \tfrac{1}{2}\min\Big(1, \frac{(T-250)^+}{250}\Big) - \tfrac{1}{2}\min\Big(1, \frac{(E-50)^+}{50}\Big),$$

$$f_{\text{ada}} = \min\Big(1, \frac{n_{\text{ADA}}}{n_{224.2}}\Big) \cdot \min\Big(1, \frac{\text{floors with ADA}}{\text{target}}\Big), \qquad f_{\text{core}} = \text{mean}_i\ \Big(1 - \frac{\min(\delta_i, 120)}{120}\Big),$$

$$f_{\text{adj}} = \frac{\sum_j \omega_j \max\big(0,\ 1 - (\Delta_j - m_j)^+ / m_j\big)}{\sum_j \omega_j}, \qquad f_{\text{corner}} = \frac{\#\{\text{corner rooms on a corner}\}}{\#\{\text{corner rooms}\}},$$

$$f_{\text{bal}} = \text{clip}\Big(1 - \frac{\text{sd}(K_\ell)}{\text{mean}(K_\ell)}\Big), \qquad f_{\text{comp}} = \text{clip}\Big(\frac{2S_{\text{shared}}}{0.6\,P_{\text{rooms}}}\Big).$$

Here a_i is the drawn area (credit is capped at the nominal area, so an oversized room is not worth more), Δ_j the distance between the two rooms of adjacency j with allowed distance m_j, K_ℓ the keys on guest floor ℓ, S_{shared} the total length of walls shared by two rooms and P_{rooms} the total room perimeter. The limits 250 ft (exit travel) and 50 ft (dead end) are read from RNGD's IBC rule CSV. The weights are placement 0.3, daylight 0.14, efficiency 0.1, egress 0.12, ada 0.08, core access 0.07, adjacency 0.07, corner 0.03, balance 0.05, compactness 0.04. There is no language model in F: it is arithmetic on the geometry.

Upper bounds (can the keys fit at all?). Two bounds hold for any layout, whatever the algorithm. A corridor of width w serves at most two room depths, so at least $w/(2D+w)$ of the used floor is corridor:

$$K \le K_{\text{area}} = \Big\lfloor \frac{L\,|\Omega|\,\big(1 - \frac{w}{2D+w}\big)}{\bar{a}} \Big\rfloor.$$

If every guest room needs a window, the rooms of one floor share the facade length P_f:

$$K \le K_{\text{window}} = L \Big\lfloor \frac{P_f}{\bar{f}} \Big\rfloor.$$

For a generated hotel plate with corridor length R and cores taking a frontage c_f, the bound per floor is $\lfloor (2R - c_f)/M \rfloor$. The bounds are loose (they ignore corners and the public rooms), but a request above them is impossible, and a result close to them is close to the best any search can do.

Search. With m candidate spines the decoder is driven by a subset of them, so the first stage enumerates

$$1 + \sum_{k=1}^{4} \binom{\min(m, 7)}{k} \le 1 + 7 + 21 + 35 + 35 = 99$$

spine sets (all spines, and every set of up to four of the seven best), with rooms in priority order. The second stage starts from the three best distinct layouts and repeats: generate the neighbours (switch one spine on or off: 10 moves; trim either end or shift an active spine: 4 moves each; new connector elbows; four room-order nudges), evaluate them in a seeded random order, and accept the first one with $F' > F + 10^{-6}$. When all three starts are local optima, the rest of the budget is spent on iterated local search: the best layout is "kicked" with two moves at once, improved again by the same pattern search, and kept only if it beats the best so far. The total budget is PG evaluations (P = population, G = generations, kept as the budget controls), the same as the GA's. The genetic algorithm (variant C) evaluates $P(G+1)$ layouts plus local-search steps, here about 804. Crossover is not used in the systematic search: splicing the corridor genes of one plan with the room order of another seldom gives a better plan, because the value of a corridor depends on which other corridors exist.

Plate generation (hotel). For K keys on F guest floors the busiest level needs net area A; with slack $\varsigma = 1.04$, core area A_c and packing efficiency $\eta = 0.93$ the corridor length to provide is

$$R = \frac{\varsigma A + A_c}{2D\eta}.$$

Each shape with bounding box $a \times b$ and wing depth $W_d = 2D + w$ (58 ft) offers $R_{\text{Bar}} = a$, $R_{\text{L}} = R_{\text{T}} = a + b - W_d$, $R_{\text{U}} = a + 2b - 2W_d$, $R_{\text{H}} = a + 2b - 2W_d$ and $R_{\text{Courtyard}} = 2a + 2b - 4W_d$; the generator solves $R_{\text{shape}}(a, \beta a) \geq R$ for the smallest a at aspect ratios $\beta \in \{0.6, 1\}$ and clips the result to the buildable area in both orientations. The elevator count is $e = \max(2, \lceil K/75 \rceil)$, and the two stairs of the "centre core" scheme are the pair of wing-end positions with the largest distance between them (exit separation, short dead ends).

Decision. Each layout j has a vector of twelve criteria $x_{j1..12}$ (keys, gross SF per key, facade per key, height, corridor share, dead end, code checks passed and others). Layout j is *dominated* if another layout is at least as good on every criterion and strictly better on one; the rest form the Pareto front. TOPSIS then ranks all layouts:

$$r_{jq} = \frac{x_{jq}}{\sqrt{\sum_j x_{jq}^2}}, \quad v_{jq} = \frac{\lambda_q}{\sum_q \lambda_q} r_{jq}, \quad d_j^{\pm} = \sqrt{\sum_q (v_{jq} - v_q^{\pm})^2}, \quad C_j = \frac{d_j^-}{d_j^+ + d_j^-} \in [0, 1],$$

where v_q^+ is the best and v_q^- the worst value of criterion q over all layouts and λ_q the weight from the chosen preset (Balanced, Most keys, Lowest cost, Guest experience, Low-rise) or the sliders. The layout with the largest C_j is ranked first.

9.5 Results on the three QA sites

All variants use `QA/rooms_hotel.json` (72 rooms on three levels) and are scored by the same evaluator. Variants B, C and D were run with population 24 and 30 generations as the budget, seed 7, no LLM. For A the search time is not stored in the files; its history holds 500 generations.

Table 22: Reference GA and the three later variants. "No window" counts guest rooms without a facade wall; "Unused" is the share of placeable area that is neither room nor corridor; "Checks" are the code checks passed.

Site	Variant	Rooms	No window	Fitness	Corridor	Unused	Dead end ft	Checks	Evaluated	Time
fp1	A Reference GA	61/72	7	0.880	0.151	0.16	33	6/6	500 gen.	n/a
	B GA, free-form	53/72	3	0.880	0.100	0.29	55	5/6	804	119 s
	C Updated GA	72/72	0	0.954	0.101	0.09	27	6/6	804	24 s
	D Systematic	72/72	0	0.950	0.101	0.09	25	6/6	695	20 s
fp2	A Reference GA	53/72	8	0.860	0.117	0.26	31	6/6	500 gen.	n/a
	B GA, free-form	60/72	8	0.893	0.111	0.19	36	6/6	804	102 s
	C Updated GA	62/72	0	0.908	0.146	0.06	43	6/6	804	22 s
	D Systematic	60/72	0	0.904	0.145	0.09	46	6/6	637	17 s
fp3	A Reference GA	57/72	9	0.851	0.157	0.18	41	6/6	500 gen.	n/a
	B GA, free-form	61/72	3	0.894	0.120	0.18	33	6/6	804	115 s
	C Updated GA	63/72	0	0.921	0.122	0.11	35	6/6	804	26 s
	D Systematic	60/72	0	0.911	0.119	0.09	29	6/6	683	21 s

Table 24: Upper bounds on the number of keys (Section 9.4) against what was asked and placed.

Site	Placeable SF / level	Program SF	Program / placeable	Facade ft / level	Mean key frontage ft	K_{area}	K_{window}	Keys asked	Keys A	Keys D
fp1	9,640	23,310	0.81	489	12.9	80	111	69	58	69
fp2	9,446	23,310	0.82	410	12.9	78	93	69	50	57
fp3	9,412	23,310	0.83	519	12.9	78	120	69	54	57

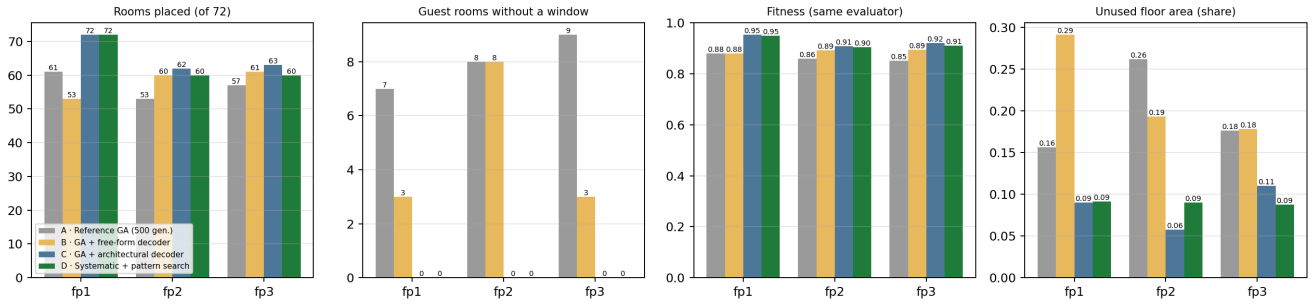


Figure 40: The four variants on the three sites (same program, same evaluator).

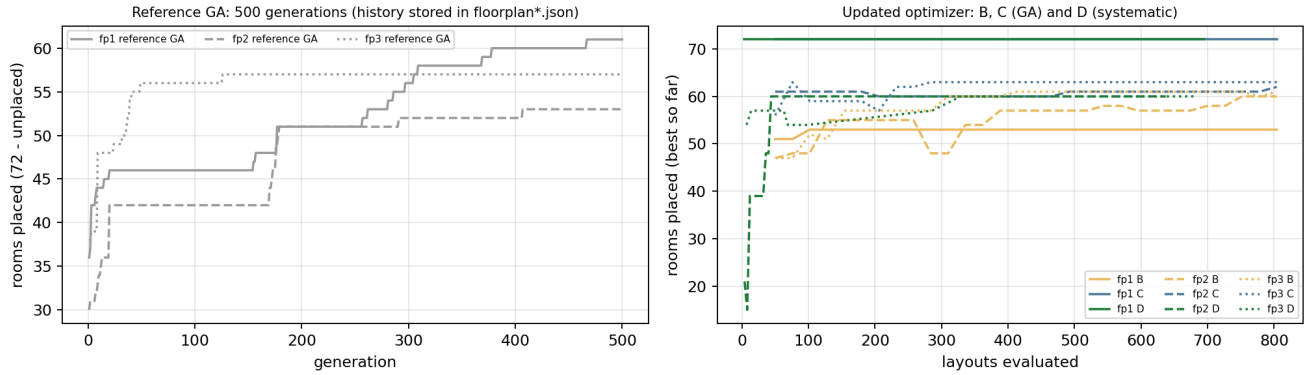


Figure 41: Left: rooms placed by the reference GA over its generations (from the history in the delivered files). Right: best rooms placed against layouts evaluated for the later variants.



Figure 42: Site fp1, Level 2, same program and same evaluator. Blue lines are window walls; a red dashed outline marks a guest room with no window; hatched bays are unused.



Figure 43: Site fp2, Level 2, same program and same evaluator. Blue lines are window walls; a red dashed outline marks a guest room with no window; hatched bays are unused.



Figure 44: Site fp3, Level 2, same program and same evaluator. Blue lines are window walls; a red dashed outline marks a guest room with no window; hatched bays are unused.

9.6 Results in Hotel Schematic (generative shape, core and floors)

Input: the Project Renegade Living + Lodging program (117 keys) and a buildable area of 170 x 265 ft. The search screened 158 plate and floor-count variants, evaluated 1,140 layouts and kept 740 distinct ones in 36 s on eight processes, with no LLM calls; a second run returned the same list: yes. 212 layouts are on the Pareto front. Pattern search improved the screened plates as follows: Courtyard 156x156 · center core (5 fl.) 0.870 → 0.870; L 202x121 · ends core (5 fl.) 0.863 → 0.881; Bar 265x58 · center core (5 fl.) 0.863 → 0.863; T 202x121 · center core (5 fl.) 0.856 → 0.865.

Table 26: Top ten layouts under the Balanced decision weights.

#	Layout	Guest floors	TOPSIS C_j	Pareto	Keys	Fitness	Corridor	SF / key	Height ft	Dead end ft	Code
1	Bar 265x58 · center core	4	0.917	yes	117	0.859	0.112	657	59	24	1.00
2	L 220x132 · knuckle core	4	0.906	yes	117	0.849	0.120	729	59	26	1.00
3	L 170x176 · knuckle core	4	0.905	yes	117	0.835	0.120	714	59	24	1.00
4	Bar 265x58 · ends core	4	0.904	yes	117	0.845	0.113	657	59	30	1.00
5	Bar 265x58 · ends core	4	0.903	yes	117	0.843	0.113	657	59	30	1.00
6	Bar 265x58 · ends core	4	0.903	yes	117	0.842	0.113	657	59	31	1.00
7	Courtyard 156x156 · ends core	4	0.902	yes	117	0.845	0.111	972	59	28	1.00
8	T 170x176 · knuckle core	4	0.899	no	114	0.840	0.122	733	59	24	1.00
9	T 170x176 · knuckle core	4	0.899	no	114	0.840	0.122	733	59	24	1.00
10	Bar 265x58 · center core	5	0.897	yes	117	0.863	0.112	788	70	24	0.86

Table 28: Best layout of each shape (Balanced weights).

Shape	Best layout of that shape	Guest floors	Overall rank	Keys	SF / key	Facade ft / key	Dead end ft	Code
Bar	Bar 265x58 · center core	4	1	117	657	27.6	24	1.00
L	L 220x132 · knuckle core	4	2	117	729	30.1	26	1.00
Courtyard	Courtyard 156x156 · ends core	4	7	117	972	33.5	28	1.00
T	T 170x176 · knuckle core	4	8	114	733	30.4	24	1.00
U	U 187x112 · center core	4	34	111	771	31.8	29	1.00
H	H 156x136 · knuckle core	4	206	106	854	34.9	59	0.86

Table 30: The first-ranked layout changes with the decision priority: the trade-off is the user's, the search only supplies the options.

Decision priority	Layout ranked first	Guest floors	Keys	SF / key	Height ft	Code
Balanced	Bar 265x58 · center core	4	117	657	59	1.00
Most keys	Courtyard 156x156 · center core	4	117	972	59	1.00
Lowest cost	Bar 265x58 · center core	4	117	657	59	1.00
Guest experience	Bar 265x58 · center core	4	117	657	59	1.00
Low-rise	Bar 265x58 · center core	4	117	657	59	1.00

First-ranked layout: Bar 265x58 · center core, 4 guest floors. Footprint 265 x 58 ft (15,370 SF), 2 elevators, bay module 13 ft. It places 117 of 117 keys and 142 of 150 program rooms; the frontage bound is 34 keys per typical floor. Corridor share 0.112, longest exit travel 64 ft, dead end 24 ft, fitness 0.859. Not placed: White Box Retail x1, IT x1, Jan x2, Gym x1, MEP Service x1, Loading x1, Luggage x1.

Rule	Check	Required	Provided	Status
IBC-R1-015	Exits per story (cores reached)	$\geq$ 2	3	**PASS**
IBC-R1-018	Exit separation (core to core)	$\geq$ 1/3 diagonal = 90 ft	202 ft	**PASS**
IBC-R1-023	Exit access travel	$\leq$ 250 ft	63 ft	**PASS**
IBC-R1-025	Corridor width	$\geq$ 44 in	72 in	**PASS**
IBC-R1-027	Dead-end corridor	$\leq$ 50 ft	24 ft	**PASS**
ADA-224.2	ADA mobility rooms	$\geq$ 7 for 117 keys	7	**PASS**
IBC-R1-007	Stories above grade	$\leq$ 5 (Type IIA, sprinklered R-1)	5	**PASS**

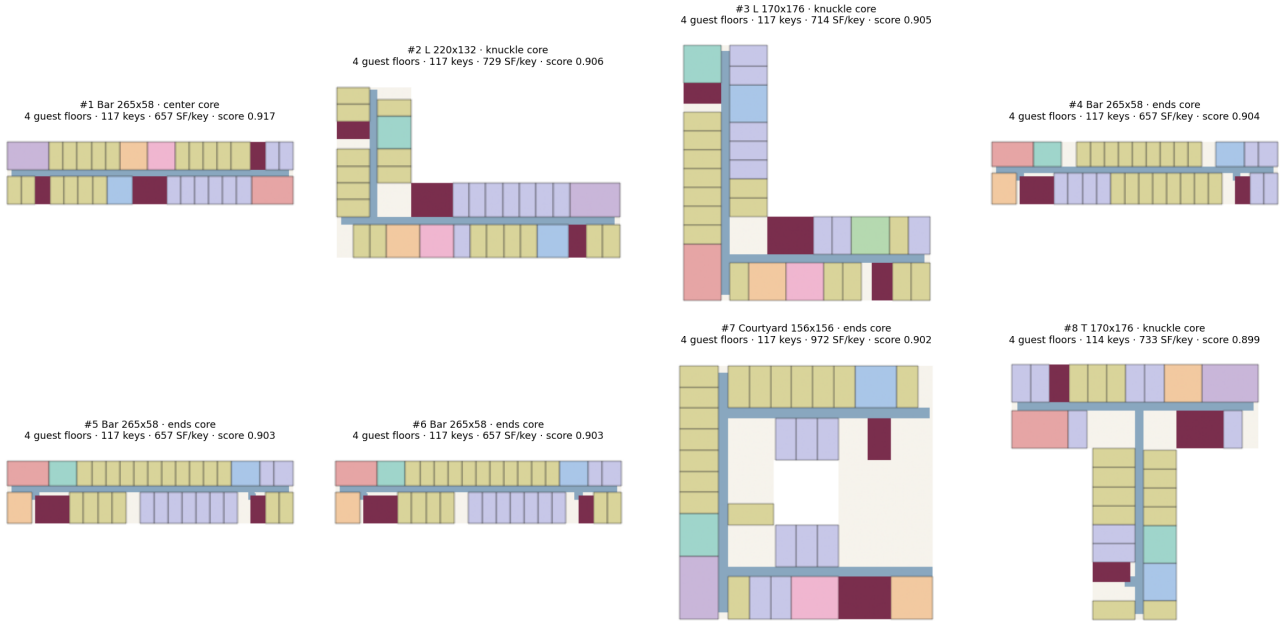


Figure 45: The eight first-ranked layouts (typical guest floor). Dark red: core and stairs; blue: corridor.

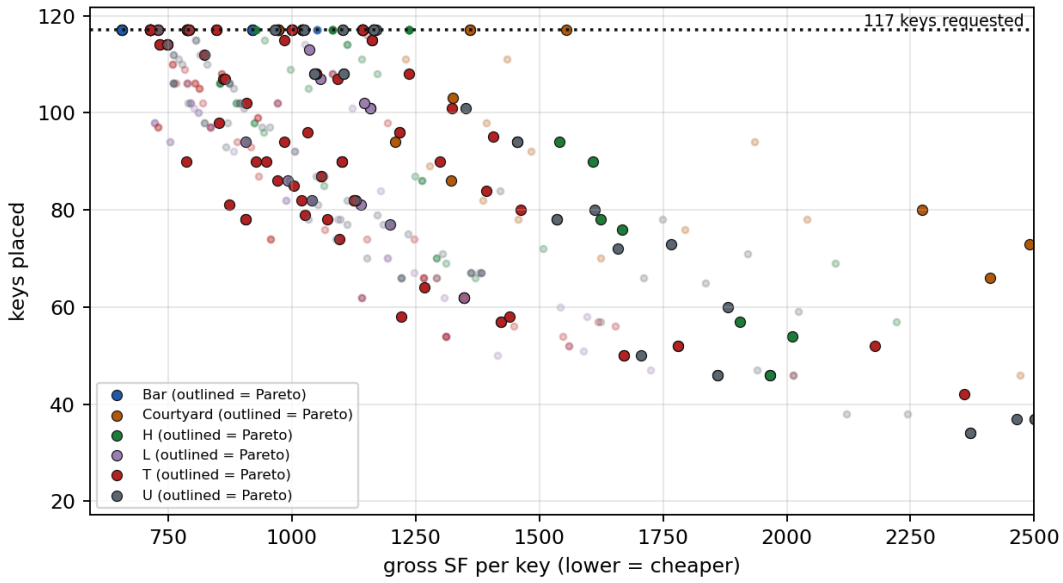


Figure 46: Every layout found: keys placed against gross area per key. Outlined points are on the Pareto front.



Figure 47: First-ranked layout, Level 2, as drawn in the app: furnished rooms, door swings, windows (blue), stairs and elevators.



Figure 48: First-ranked layout, Level 1 (public and back-of-house rooms).

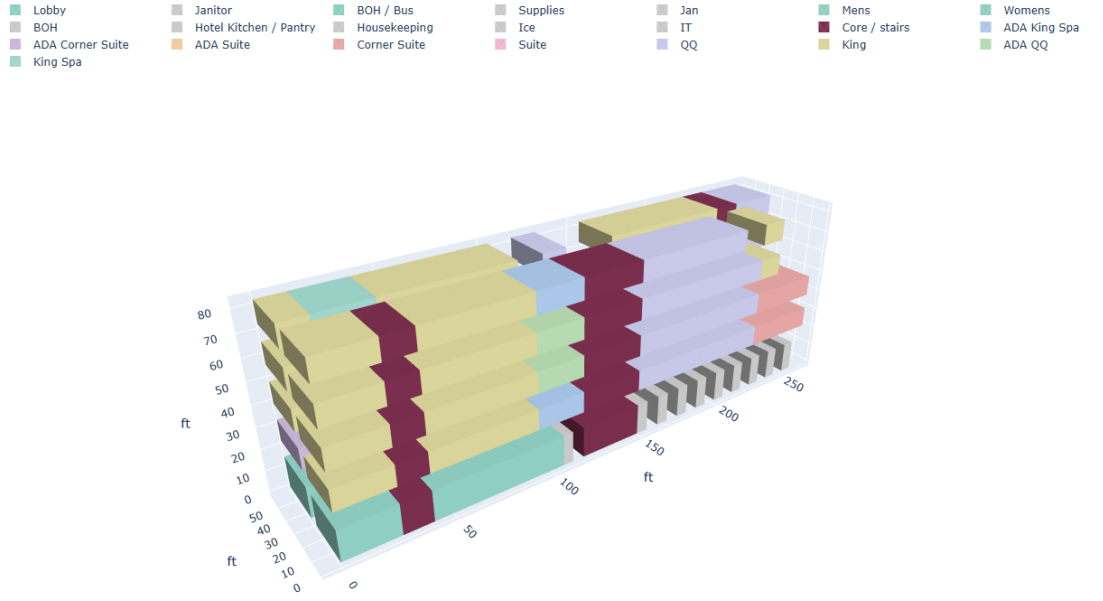


Figure 49: First-ranked layout, stacked massing with the floors pulled apart.

9.7 App screenshots

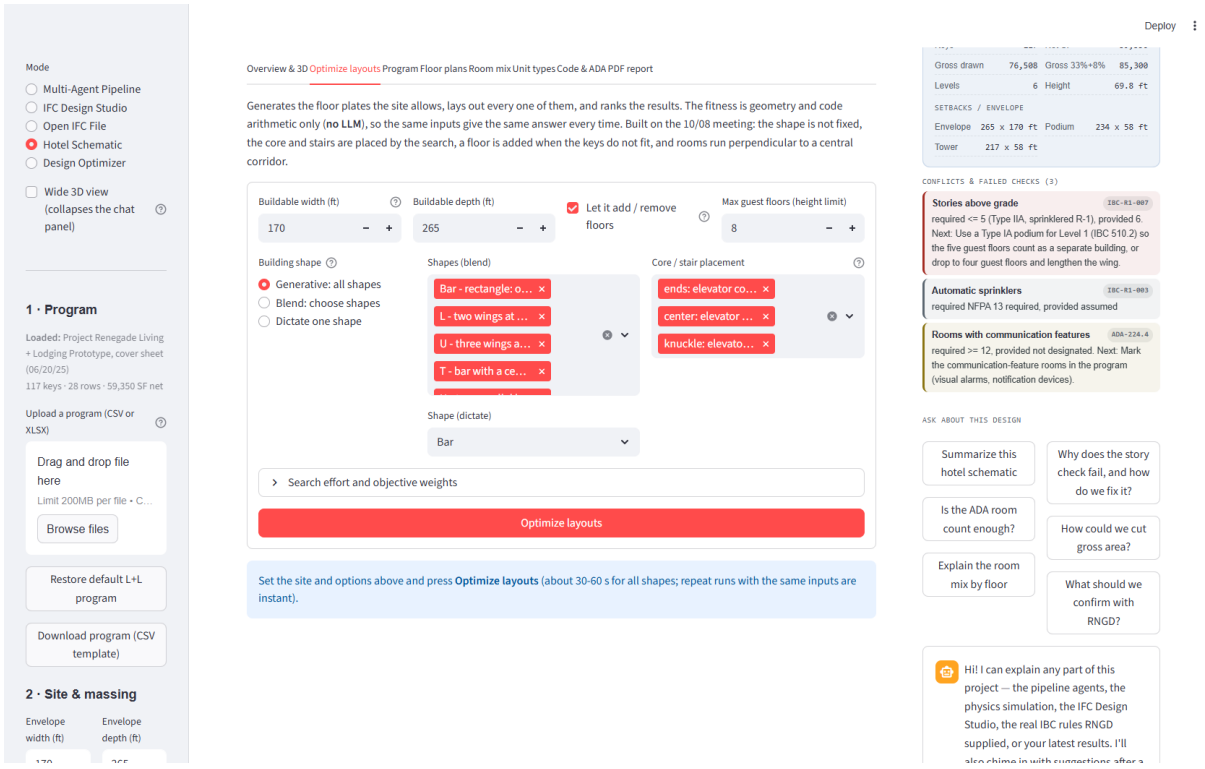


Figure 50: Optimize layouts: buildable area, shape mode (generative, blend or one shape), core placement and floors.

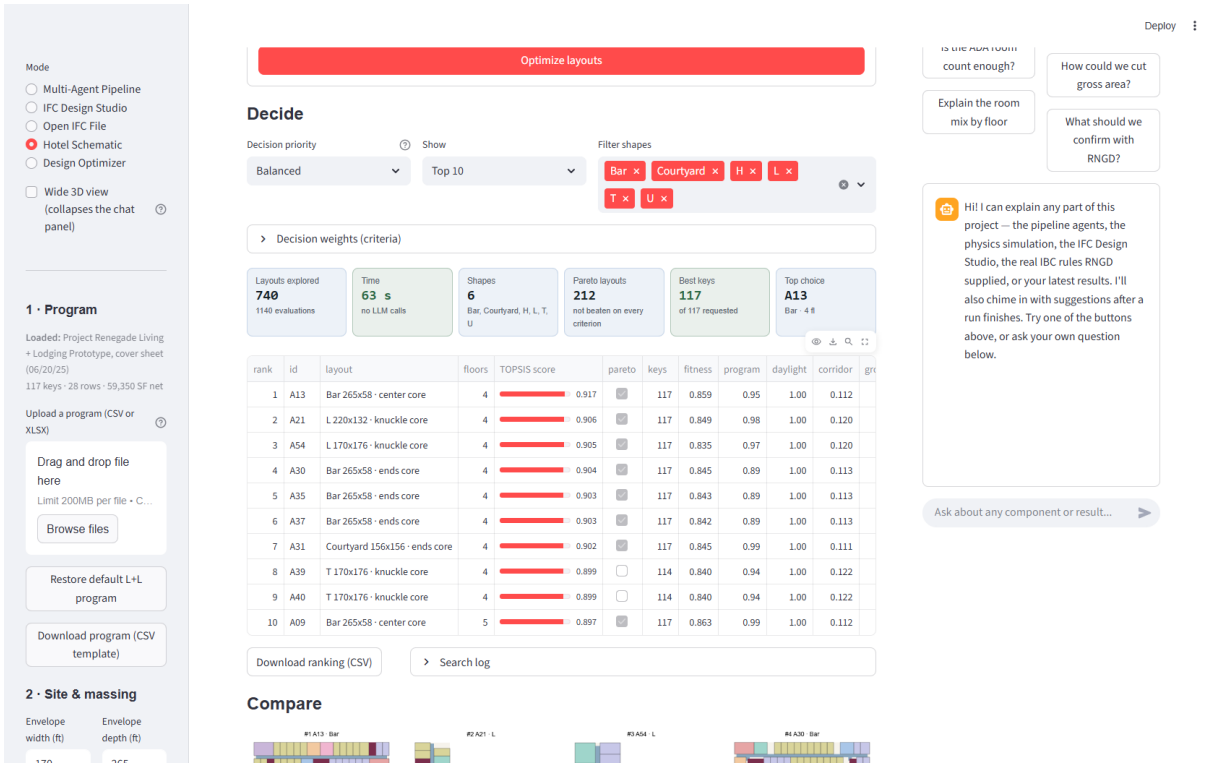


Figure 51: Decision step: every distinct layout ranked by TOPSIS under the chosen priority, with the Pareto flag; the Show dropdown switches between the top 5, 10 or 25, the Pareto front and all layouts.



Figure 52: Comparison: typical-floor thumbnails of the first-ranked layouts, keys against gross area per key for every layout, and the fitness gained by pattern search.

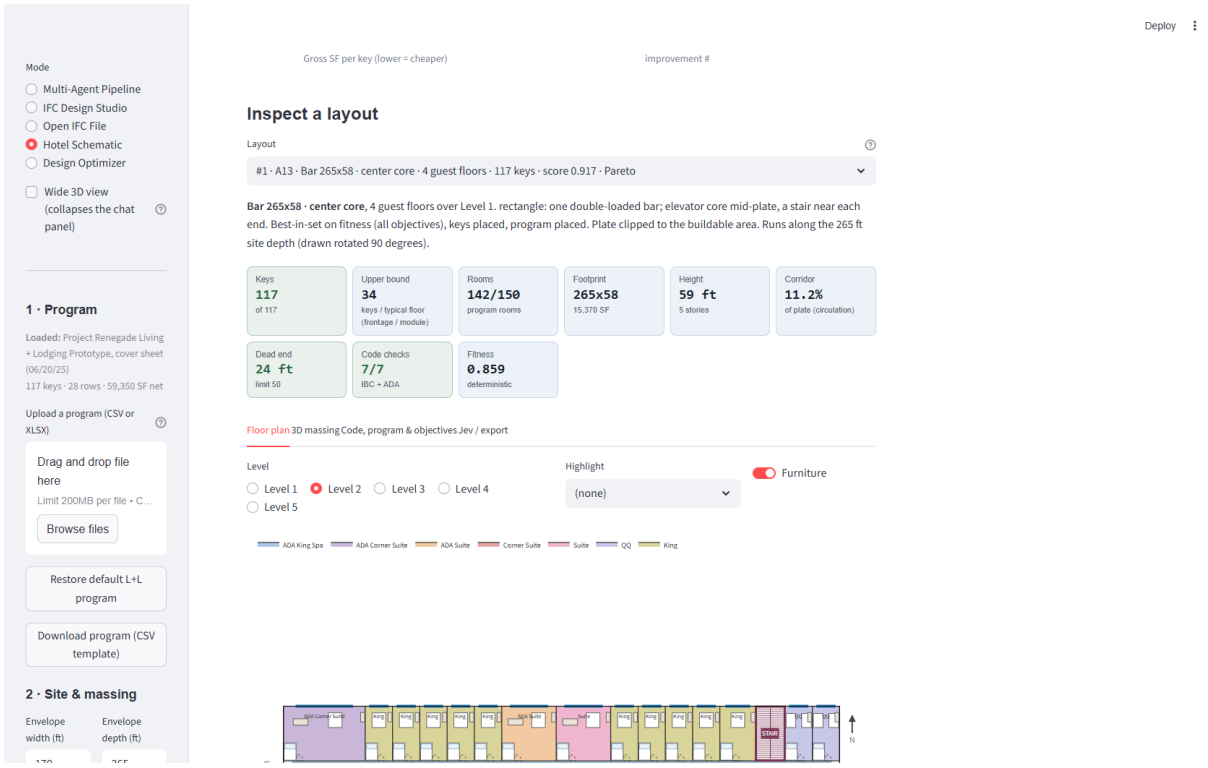


Figure 53: One layout inspected: keys, upper bound, footprint, corridor share, dead end, code checks and fitness.

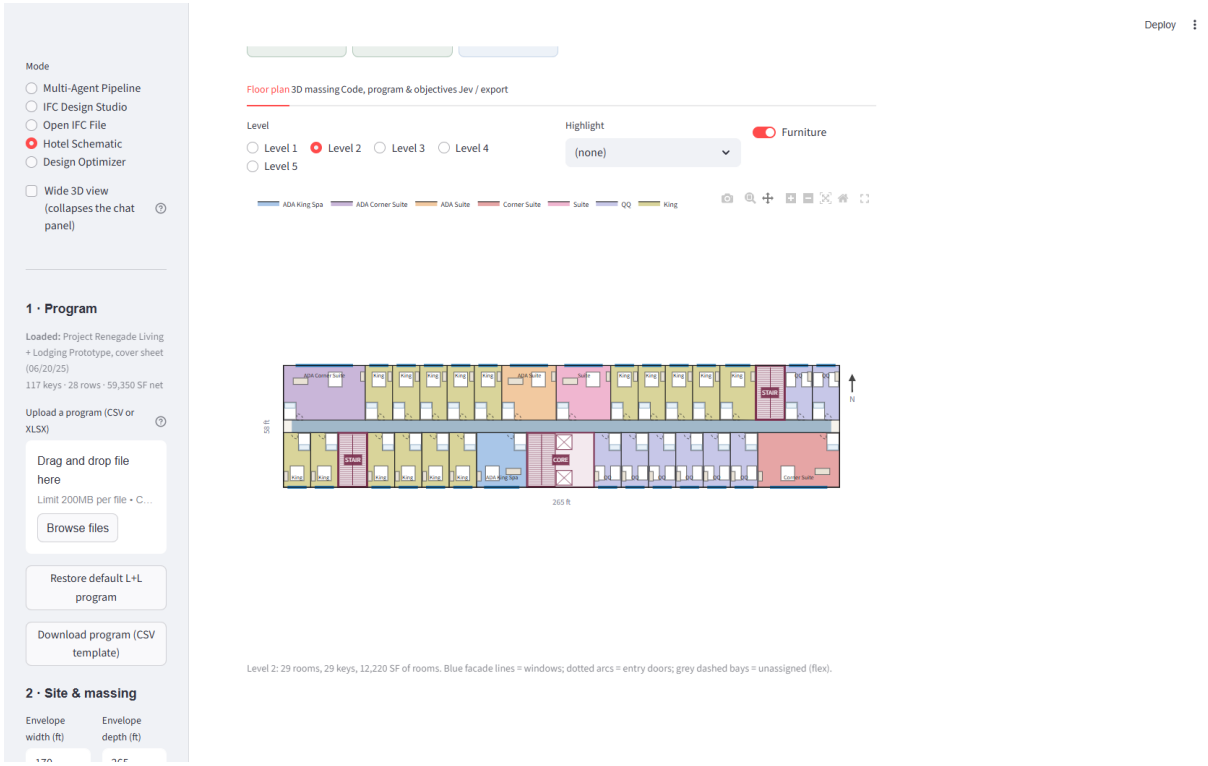


Figure 54: Interactive furnished plan of the selected layout (hover gives room data).



Figure 55: Stacked 3D massing of the selected layout, coloured by room type.

9.8 What is still open

- Level 1 uses the same plate as the guest floors (no wider podium), so large ground-floor rooms can be left out; they are listed for every layout.
- The decoder draws rectangles on a 1 ft grid: angled wings, curved facades and structural grids are not modelled.
- The upper bounds are loose; a tight bound would need an exact packing model.
- The centre-core scheme uses two stairs; on T, U and H plates the third wing can exceed the dead-end limit. The check flags it and the ranking moves such layouts down.
- Parking, cost and energy are not in the fitness. Facade length per key and gross area per key stand in for cost in the decision step.
- Jev (TypeSafe's hosted decision model) is wired only as an optional advisor; no key was configured for this run.

10 Assistant

The chat panel answers with the live state of whichever mode is open (pipeline result, IFC design, real IFC summary, hotel schematic) plus retrieved passages from the knowledge base and RNGD's documents.

Question (one-click button): Why does the story check fail, and how do we fix it?

Answer, as given in the run (reproduced verbatim; LLM answers are not fact-checked and can misstate code details — the code screen in Section 8 is the authoritative result):

> Why the story-count check fails

Rule	Requirement	What the model shows	Why it fails

IBC-R1-007 – Stories above grade (RNGD's real 2021 IBC CSV)	$\leq$ 5 stories above the grade plane for a sprinklered R-1, Type IIA building	6 stories above grade (Level 1 podium + 5 guest floors)	The podium is being counted as a "story above grade" because it is a finished, occupiable floor (lobby/BOH). The IBC treats any occupiable floor that meets the definition of a story (height $\geq$ 8 ft, has a ceiling, is not a mezzanine) as part of the story count. With the podium included, the total exceeds the 5-story limit.

How to bring the design into compliance

Re-classify the podium

If the podium can be treated as a non-habitable "parking/mechanical" level (Type IA, not Type IIA), the IBC allows it to be excluded from the "stories above grade" count.

Change the podium program from Lobby + BOH to a parking/garage or service level that is not occupied (no sleeping rooms, no assembly).

Update the code-screen to mark the podium as "non-habitable"; the story count will then be 5 (guest floors only) and the rule will PASS.

Reduce the number of guest floors

Drop one guest floor (e.g., go from 5 guest floors to 4).

Keep the podium as a regular lobby/BOH (still counted) → total stories = 5 → PASS.

This will lower total GSF and may affect unit count; you'll need to re-run the BaaP matching to see if the 117-key program still fits.

Apply for a zoning/IBC variance

If the client insists on keeping the podium as an occupied floor and the 5-story limit, a variance from the local building authority can be requested.

Document the rationale (e.g., podium provides essential public amenities, no additional fire-risk, meets fire-rating requirements).

Note: The variance is a non-technical fix; the pipeline will still flag the rule as FAIL until the variance is granted.

Combine (1) + (2) for flexibility

Convert the podium to a parking/garage level (non-habitable) and keep 5 guest floors.

This satisfies the story count and may help the parking shortfall (see the earlier "parking-shortfall" issue).

What to do next in the tool

Edit the hotel schematic (in the "Hotel Schematic" UI) → change the Level 1 program type or delete one guest floor.

Re-run the "Code screen" (the real-IBC check tab) to verify that IBC-R1-007 now PASSes.

If you keep the podium as a lobby, run the IFC Design Studio with a structured-parking garage under the podium; that may also solve the separate parking-stall shortfall.

Bottom line: The failure is purely a code-count issue—reduce the counted stories or re-classify the podium so it no longer counts as an "above-grade" story. Once that is done, the story-count rule will pass, and you can focus on the remaining parking conflict.

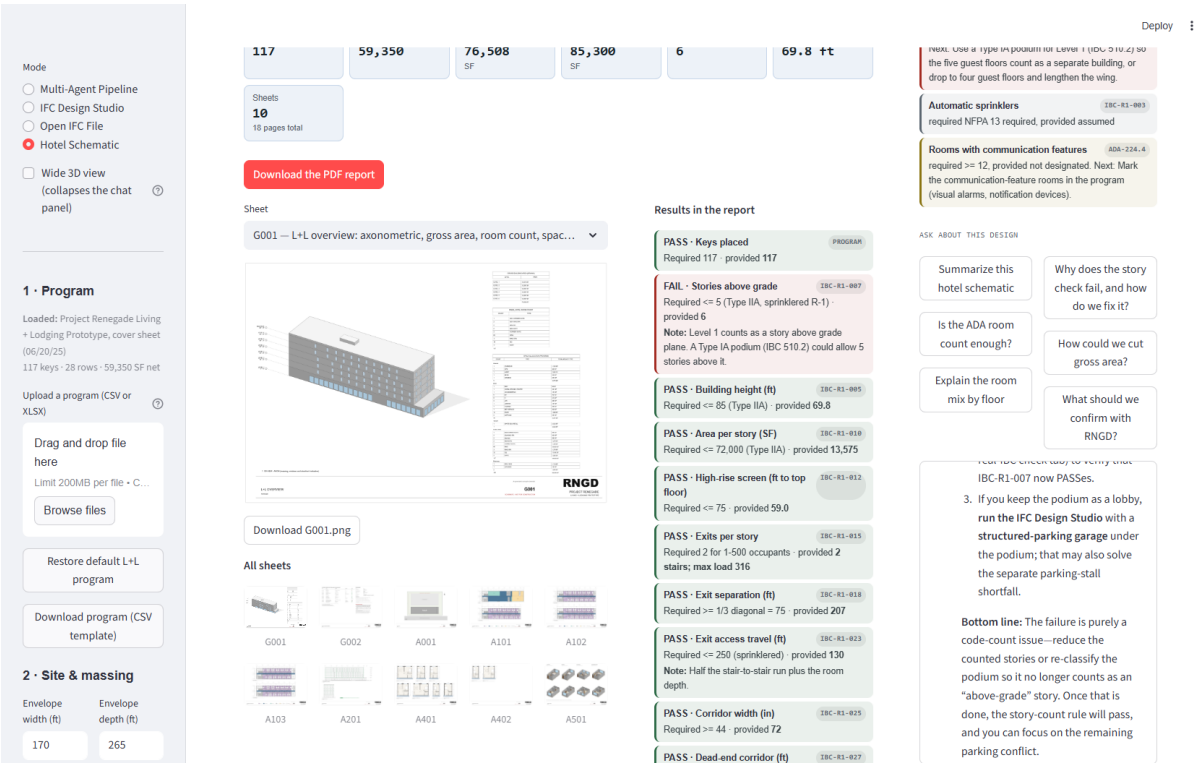


Figure 56: The assistant answering a one-click question about the hotel design, using the live design state.

11 Verification: automated tests

The full pytest suite was run as part of building this report: 51 of 51 tests passed (51 passed (results of the last full run, pytest_results.xml)). App tests drive the real Streamlit app through Streamlit's AppTest, including a full pipeline run with real LLM calls.

File	Test	s	Result
test_app	test_loads_without_exception_default_mode	28.2	PASS
test_app	test_ifc_studio_scenario_button_generates_design	4.4	PASS
test_app	test_ifc_studio_shows_real_ibc_tab_and_summary_card	1.6	PASS
test_app	test_quick_question_button_asks_and_replies	32.6	PASS
test_app	test_real_docs_loaded_into_knowledge_base	0.1	PASS
test_app	test_chat_panel_replies	20.2	PASS
test_app	test_open_ifc_file_explorer	9.4	PASS
test_app	test_full_pipeline_run_via_ui	52.2	PASS
test_app	test_hotel_mode_tabs_and_controls	6.7	PASS
test_app	test_hotel_optimize_layouts_tab	13.4	PASS
test_app	test_design_optimizer_mode_runs_and_exports_pdf	5.6	PASS
test_hotel	test_default_program_totals	0.0	PASS
test_hotel	test_net_to_gross_rounds_up_to_hundred	0.0	PASS
test_hotel	test_schematic_places_all_keys	0.0	PASS
test_hotel	test_six_levels_flagged_against_five_story_rule	0.0	PASS
test_hotel	test_upload_csv	0.0	PASS
test_hotel	test_detailed_program_matches_source_schedule	0.0	PASS
test_hotel	test_stairs_and_core_stack_on_every_level	0.0	PASS
test_hotel	test_per_floor_support_rooms_on_each_guest_floor	0.0	PASS
test_hotel	test_two_end_stairs_clear_dead_end_and_fit_envelope	0.0	PASS

File	Test	s	Result
test_hotel	test_ground_floor_places_every_non_guest_row	0.0	PASS
test_hotel	test_uploaded_program_draws_a_plan_sheet	0.9	PASS
test_hotel	test_ada_tables_match_2010_standards	0.0	PASS
test_hotel	test_program_detail_and_mix_by_floor	0.0	PASS
test_hotel	test_per_floor_rooms_follow_floor_count	0.0	PASS
test_hotel	test_interactive_figures_build	0.1	PASS
test_hotel_optimize	test_plates_fit_envelope_and_have_cores	0.3	PASS
test_hotel_optimize	test_elevators_follow_keys	0.0	PASS
test_hotel_optimize	test_explore_is_deterministic_and_places_the_keys	4.8	PASS
test_hotel_optimize	test_decision_ranking_pareto_and_topsis	2.6	PASS
test_ifc_design	test_module_counts_sum_to_units	0.0	PASS
test_ifc_design	test_compliant_scenario_passes_all_checks	0.0	PASS
test_ifc_design	test_baseline_reports_parking_shortfall	0.0	PASS
test_ifc_design	test_higher_floor_count_shrinks_footprint	0.0	PASS
test_ifc_design	test_zoning_preset_changes_limits	0.0	PASS
test_ifc_design	test_accessible_units_present	0.0	PASS
test_ifc_design	test_ifc_roundtrip[Baseline: 4 floors, surface parking]	1.5	PASS
test_ifc_design	test_ifc_roundtrip[Right-sized: 60 units, podium (compliant)]	0.8	PASS
test_ifc_design	test_ifc_roundtrip[Podium parking, 5 floors]	1.5	PASS
test_ifc_design	test_ifc_roundtrip[Dense R-6, 7 floors + podium]	1.4	PASS
test_ifc_design	test_ifc_roundtrip[Suburban R-2, 60 units, 3 floors]	0.8	PASS
test_optimizer	test_site_and_programs_load	0.0	PASS
test_optimizer	test_reference_designs_reconstruct_exactly[1-982]	0.0	PASS
test_optimizer	test_reference_designs_reconstruct_exactly[2-737]	0.0	PASS
test_optimizer	test_reference_designs_reconstruct_exactly[3-913]	0.0	PASS
test_optimizer	test_decoder_is_valid_and_deterministic	0.1	PASS
test_optimizer	test_search_improves_and_exports	1.1	PASS
test_optimizer	test_heuristic_review_and_weight_tuning	0.0	PASS
test_optimizer	test_fact_check_flags_invented_numbers	0.0	PASS
test_optimizer	test_layout_quality_on_reference_program	0.2	PASS
test_optimizer	test_systematic_search_is_deterministic_and_monotonic	3.0	PASS

12 Limitations

- Pipeline and IFC-studio inputs (customer request, site, zoning presets, BaaP catalog) are synthetic sample data.
- LLM outputs vary between runs; the pipeline numbers above are from this run only.
- The hotel layout is a rule-based schematic: parking, structure and MEP are not modelled; fixtures are typical.
- The optimized layouts are rectangles on a 1 ft grid with one plate for all levels; open items are listed at the end of Section 9.
- Code screens are planning-level, not code determinations. 3D views are WebGL (app) and painter's-algorithm axonometrics (PDF).

A Screenshot index

File	Mode	Caption
01_pipeline_inputs.png	Multi-Agent Pipeline	Multi-Agent Pipeline before a run: the sample customer request and site sketch that feed the agents; the assistant panel on the right.

File	Mode	Caption
02_pipeline_running.png	Multi-Agent Pipeline	The pipeline running live: each agent reports its step as it finishes.
03_pipeline_overview.png	Multi-Agent Pipeline	Pipeline result overview: the arbiter's decision, confidence and conflict count.
04_pipeline_agent_trace.png	Multi-Agent Pipeline	Agent trace: every agent step with model, status and latency.
05_pipeline_program_site.png	Multi-Agent Pipeline	Program parsed from the request and the computed site constraints.
06_pipeline_baap_conflicts.png	Multi-Agent Pipeline	BaaP module counts and the conflicts the Conflict Agent found.
07_pipeline_site_simulation.png	Multi-Agent Pipeline	Physics-style site simulation and the computer-vision QA of the rendered plan.
08_pipeline_floor_plates.png	Multi-Agent Pipeline	Rule-based floor plates (ground and typical floor).
09_pipeline_final_decision.png	Multi-Agent Pipeline	The arbiter's final decision, unresolved issues and next steps.
10_ifc_baseline_3d_model.png	IFC Design Studio	IFC Design Studio, Baseline scenario (120 units, 4 floors, surface parking): the generated IFC4 model in the 3D viewer.
11_ifc_baseline_site_plan.png	IFC Design Studio	Baseline site plan: building, parking, easement and fire lane on the synthetic lot.
12_ifc_baseline_compliance.png	IFC Design Studio	Baseline compliance against the ten sample-zoning checks.
13_ifc_baseline_real_ibc_check.png	IFC Design Studio	Baseline cross-checked against RNGD's real 2021 IBC rules.
14_ifc_baseline_how_this_was_built.png	IFC Design Studio	Step-by-step explanation of how the layout was generated.
15_ifc_rightsized_compliance.png	IFC Design Studio	Right-sized scenario (60 units, podium parking): the compliance checks after the fix.
16_ifc_rightsized_3d.png	IFC Design Studio	Right-sized scenario in 3D.
17_explorer_3d_model.png	Open IFC File	A real IFC file (CDI Solutions, Metairie Bank, IFC2X3, 89 MB) recentred and shown in 3D.
18_explorer_summary.png	Open IFC File	What is in the real model: schema, storeys and element counts by IFC class.
19_explorer_element_browser.png	Open IFC File	Searchable list of the model's elements.
20_hotel_overview.png	Hotel Schematic	Hotel Schematic workspace: KPI tiles, the room-by-room 3D model and the key results with suggested fixes.
21_hotel_program.png	Hotel Schematic	The editable program (Project Renegade L+L space allocation, 28 rows) with area and room-mix charts.
22_hotel_floor.png	Hotel Schematic	Interactive floor plan of Level 1 (hover shows each room's area and size).
23_hotel_program_detail.png	Hotel Schematic	Detailed program: per-room size, beds, sleeping capacity, placement, IBC occupant load and fit-out.
24_hotel_plan_level2.png	Hotel Schematic	Interactive plan of Level 2, a typical guest floor (24 keys).
25_hotel_plan_drawing.png	Hotel Schematic	The same level in drawing style (as used on the PDF sheets).
26_hotel_room.png	Hotel Schematic	Keys of each room type on each guest floor.
27_hotel_unit.png	Hotel Schematic	Unit type detail: enlarged plan and two 3D interior views.
28_hotel_code.png	Hotel Schematic	Code screen against RNGD's IBC rule set and the 2010 ADA guest-room tables, with rule text.
29_hotel_report_results.png	Hotel Schematic	PDF report tab after generating: summary numbers and the results that went into the report.
30_hotel_report_sheets.png	Hotel Schematic	Sheet gallery of the generated drawing set.
31_assistant_answer.png	Assistant	The assistant answering a one-click question about the hotel design, using the live design state.